

H2GUARD · BUILD PLAN

H2Guard prototype build plan

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-02	Amish Chadha	Decisions of 2026-10-02 carried into the plan: head on a ceiling drop rod (new section 3.4, step 4 rewritten), pressure switch (step 11), dry-contact relay, clear-lid box with eight glands, hydrogen-rated valve, oxygen sensor option (step 15); pictures regenerated
0.2	2026-10-02	Amish Chadha	Safety stop S5: head placement, airflow proving, valve, oxygen sensor and set point conditions decided on 2026-10-02 (HGD-DEC-001)
0.1	2026-10-01	Amish Chadha	First build plan, with pictures by component and step; design made constructable (HGD-DDR-003)

H2Guard prototype build plan

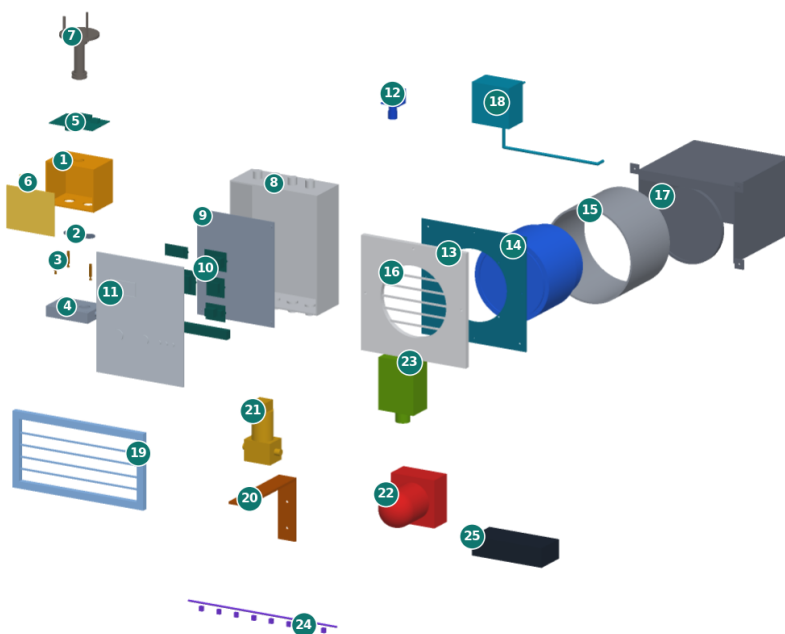
Plan, not yet built. How to build the first proof-of-concept prototype, component by component. Building and testing to it is TRL 4 work. Decisions still to be made are kept in the design decisions register ([docs/06-design-decisions.md](#)), not here.

1. What you are building

H2Guard prototype: every component, pulled apart

Numbered in build order. Seen from the front left and above; sizes to scale with each other

- 1 Detector head box, drilled
- 2 Flame arrestor discs (2)
- 3 Standoffs and M3 screws (4 + 8)
- 4 Bump test cup, printed, with push-in fitting
- 5 Sensor board with both sensors
- 6 Head lid and cable gland
- 7 Ceiling drop rod: flange, pipe, locknuts
- 8 Controller box, drilled, with 8 glands
- 9 Controller mounting plate
- 10 Controller modules, dry-contact relay, terminals
- 11 Controller lid with front panel parts
- 12 Test port bracket and test port
- 13 Fan plate
- 14 Exhaust fan
- 15 Wall sleeve, 200 mm
- 16 Inside grille
- 17 Backdraft shutter and weather hood
- 18 Pressure switch and its tube
- 19 Make-up air grille
- 20 Valve bracket
- 21 Normally closed solenoid valve
- 22 Sounder and beacon
- 23 Oxygen sensor (option)
- 24 Bump test tube (drawn short) and clips
- 25 24 V power supply



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github.com/BoujeeEnjinia1701/h2guard

Figure 1. Every component pulled apart and numbered in build order. The tube (24) is drawn short; it is about 3.1 m long.

The prototype is one H2Guard installed in a teaching room of about 30 m³ (4 x 3 x 2.5 m): a small detector head hanging from the ceiling on a short drop rod directly over the bench apparatus, a controller box with a clear lid at chest height, an exhaust fan high in the back wall with a pressure switch beside it that proves the air is moving, a make-up air grille low on the far side wall, a normally closed solenoid valve rated for hydrogen on the supply, a sounder and beacon by the door, a plug-in 24 V supply, and a thin tube that carries test gas from a port beside the controller up to the head. Where inert gas cylinders share the room, an oxygen sensor is added at breathing height. A relay in the controller gives a volt-free contact that opens on a trip or loss of power, so a host bench can cut its own supply. Figure 1 shows the 25 components in the order you make or fit them. Nine are made or drilled in a small

workshop: the head box, the printed bump test cup, the sensor board, the controller box and its lid, the controller mounting plate, the test port bracket, the fan plate and the valve bracket. Everything else is bought and fitted. The work is drilling plastic boxes, cutting and drilling aluminium sheet, angle and flat bar, bending one bar, one 3D print, soldering two sensors to a prototype board and wiring bought modules with screw terminals. The gas connection to the valve is made by a competent gas fitter, and the two wall openings for the fan and the make-up air grille are builder's work. The parts cost about USD 411 from the bill of materials, without the oxygen sensor option.

Safety: Hydrogen is flammable in air from about 4 % to 74 % by volume and ignites very easily. No hydrogen is used while building; the first gas, a certified 1 % test mixture, comes in only at the safety stops of section 6. H2Guard is a research and teaching prototype, not a certified gas detection system, and must not be the only safeguard in a room where certified detection is required. The 24 V supply is a bought, certified plug-in unit: no mains wiring is part of this build.

■ 2. What changed to make it buildable

The concept showed what H2Guard does and where each part goes in the room; some of its parts could not be made, fitted or held as drawn. Each change below keeps what the system does. The construction changes are recorded in decision record HGD-DDR-003, accepted by Amish on 2026-10-02; the last five rows come from the decisions Amish made on the same day, recorded in the design decisions register.

Table 1. Changes from the concept.

Component	The concept had	The buildable design has	Why
Bump test cup	A solid top that covered both sensor ports	Two 22 mm openings over the ports (Figure 3)	Gas can reach the sensors
Arrestor discs	25 mm discs in 26 mm ports, with nothing to hold them	22 mm ports; each disc rests on the ledge and is bonded round its edge with epoxy (Figure 3)	The disc cannot fall out and gas cannot pass round it
Head fixing	A wall plate that cut into the box, and no screws	The box hangs from the ceiling on a short drop rod: a floor flange on the ceiling, a pipe nipple and two locknuts that clamp the top of the box (Figure 9)	The ports sit directly over the leak point, inside the rising plume
Sensor board and cup	No fixing	Four 22 mm standoffs: screws from below hold the cup, screws from above hold the board (Figure 5)	One set of parts holds both and sets the sensor height
Test gas nozzle	A solid stub	A push-in fitting for 4 mm tube in a boss on the cup (Figure 6)	A standard fitting the tube pushes into
Controller inside	A board floating in the box	An aluminium mounting plate on the box's four bosses, modules on standoffs (Figure 12)	Bought modules, nothing floating
Front panel	A panel floating in front of the box	The box's own lid, cut for the display, key, button and lights (Figure 14)	The lid seals the box with the panel parts in it
Cable entries	None	Five cable glands, four on top and one underneath (Figure 10)	Keeps the box sealed
Test port	A block overlapping the port	A short aluminium angle on the wall with the port pointing down through it (Figure 16)	Test gas connects from below at chest height
Exhaust fan	A 146 mm motor floating in a 150 mm sleeve	The 170 mm fan body in a 200 mm sleeve, held by a fan plate on the inside wall (Figure 18)	The air path is still 150 mm, so the fan's delivery is unchanged
Solenoid valve	Hanging on the gas pipe, which passed through it	On a bent flat-bar bracket from the wall (Figure 21)	The gas joints carry no weight
Airflow proving	Fan rotation only	A pressure switch beside the fan, its tube ending in the fan inlet (Figure 19)	A blocked duct or stuck shutter now closes the valve
Valve body	Brass, ratings unknown	Stainless steel, rated by its maker for hydrogen with a stated seat leak rate	A leaking seat is the remaining undetected fault
Controller box	Grey lid, five glands	Clear lid, eight glands (four on top, four underneath)	Lights and relays seen without opening; room for the new cables
Interlock output	None	A dry-contact relay on the mounting plate	A host bench can cut its own supply
Oxygen	Not monitored	An oxygen sensor option at breathing height (step 15)	The catalytic sensor reads low in an oxygen-poor room

■ 3. Making the components

Make and check each component before the assembly step that needs it. Sizes are in millimetres. "Left" and "right" are as seen standing in the room facing the wall the part is on. Workshop tolerance is 0.5 mm unless a step says otherwise; drawings do not carry tolerances before TRL 4.

3.1 Detector head box, drilled

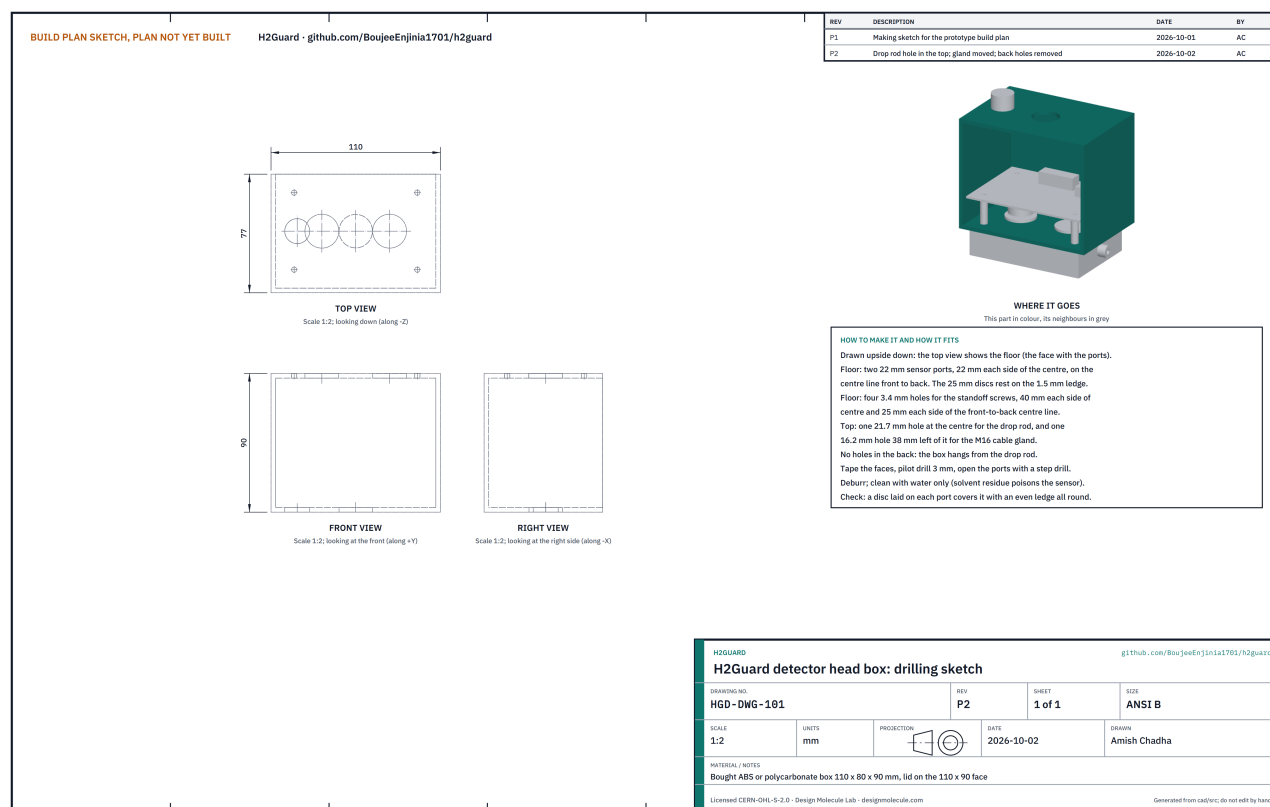


Figure 2. Detector head box drilling sketch (HGD-DWG-101), drawn upside down so the top view shows the floor.

What it is and what it is made from. The small box at the ceiling that carries both sensors. A bought ABS or polycarbonate box 110 wide, 80 deep and 90 tall, with its lid on one 110 x 90 face; the lid faces into the room.

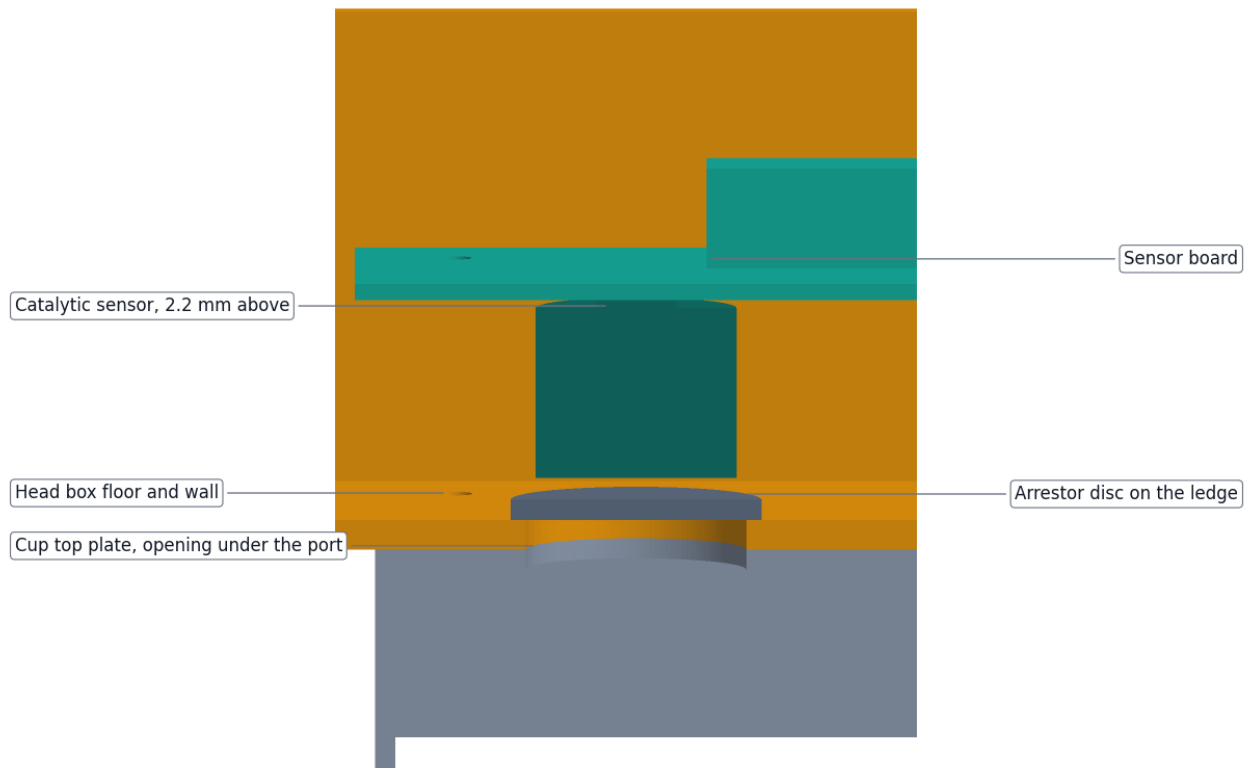
How to make it.

1. Cover the floor and top with masking tape and mark the centre lines.
2. Floor: two sensor ports, 22 mm diameter, 22 mm each side of the centre on the centre line front to back. Pilot drill 3 mm at low speed with a block of wood behind, then open with a step drill and light pressure.
3. Floor: four 3.4 mm holes for the standoff screws, 40 mm each side of the centre and 25 mm each side of the front-to-back centre line (80 x 50 mm apart).
4. Top: one 21.7 mm hole at the centre for the drop rod, and one 16.2 mm hole for the cable gland, 38 mm left of the centre on the same line. No holes go in the back.
5. Deburr inside and out. Clean with water only: solvent and flux residue poison the catalytic sensor.

How it fits the parts next to it.

Joint 1: arrestor disc, port and catalytic sensor (head cut open)

Cut through the left port, seen from the front. Gas passes the cup opening, the port and the disc to the sensor



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Figure 3. Cut through the left port. Test gas or a leak passes the opening in the cup, the 22 mm port and the 2 mm sintered disc to the sensor 2.2 mm above it.

Each 25 mm sintered stainless disc lies on the inside of the floor over its port, resting on the 1.5 mm ledge all round, and is bonded with a bead of two-part epoxy round its edge so no gas can pass round it. Never use silicone sealant anywhere in the head: it poisons the catalytic sensor. The cup screws flat to the underside of the floor and the sensor board stands on the standoffs above the discs (section 3.2). The box hangs on the drop rod (section 3.5) with its top 85 mm below the ceiling.

Check before moving on. A disc laid on each port covers it with an even ledge all round; the four standoff holes line up with the holes in the cup.

3.2 Bump test cup, printed

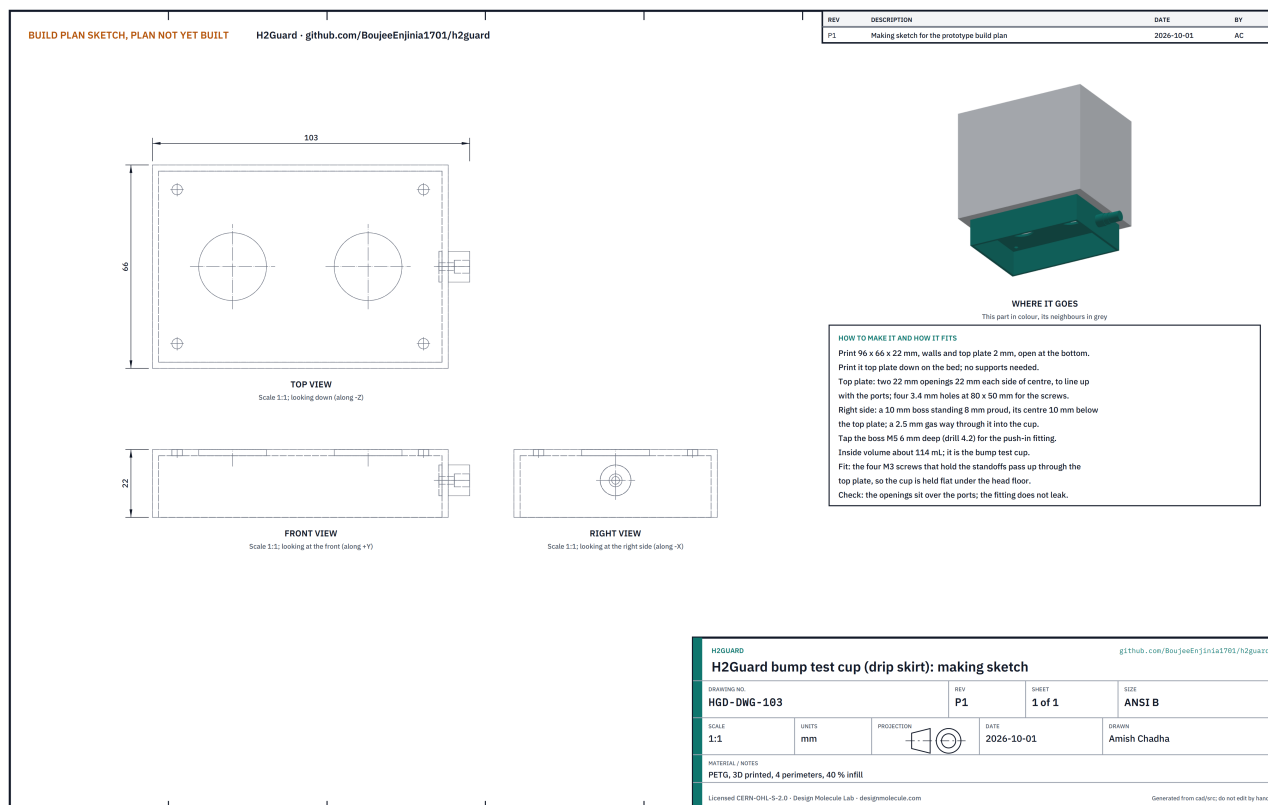


Figure 4. Bump test cup making sketch (HGD-DWG-103).

What it is and what it is made from. The open-bottomed skirt under the head. It keeps drips off the ports and, during a bump test, holds the test gas round them. PETG, printed with four perimeters and 40 % infill.

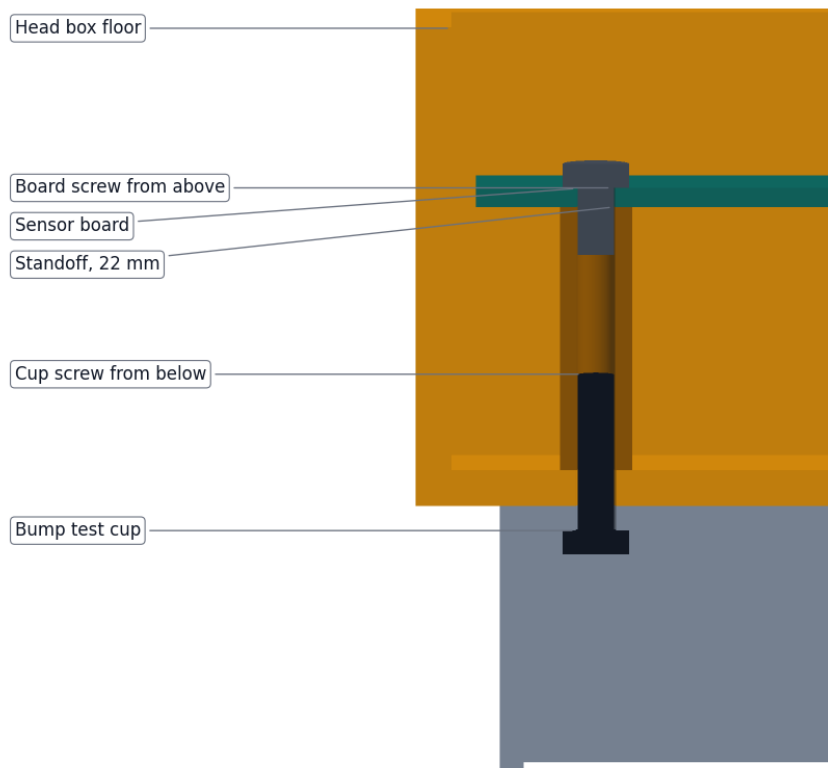
How to make it.

1. Print the cup 96 x 66 x 22 mm, walls and top plate 2 mm, top plate down on the bed; no supports are needed.
2. The print carries two 22 mm openings in the top plate, 22 mm each side of the centre, and four 3.4 mm holes 80 x 50 mm apart. Check them against the head box before going on.
3. On the right side, a boss 10 mm across stands 8 mm proud, its centre 10 mm below the top face, with a 2.5 mm gas way into the cup. Tap the boss M5 6 mm deep (drill 4.2 mm first if the printed hole is undersize).
4. Screw the M5 straight push-in fitting for 4 mm tube into the boss with thread sealant tape.

How it fits the parts next to it.

Joint 2: one standoff holds the cup and the sensor board

Cut through a standoff, seen from the front. The screw from below clamps the cup to the floor



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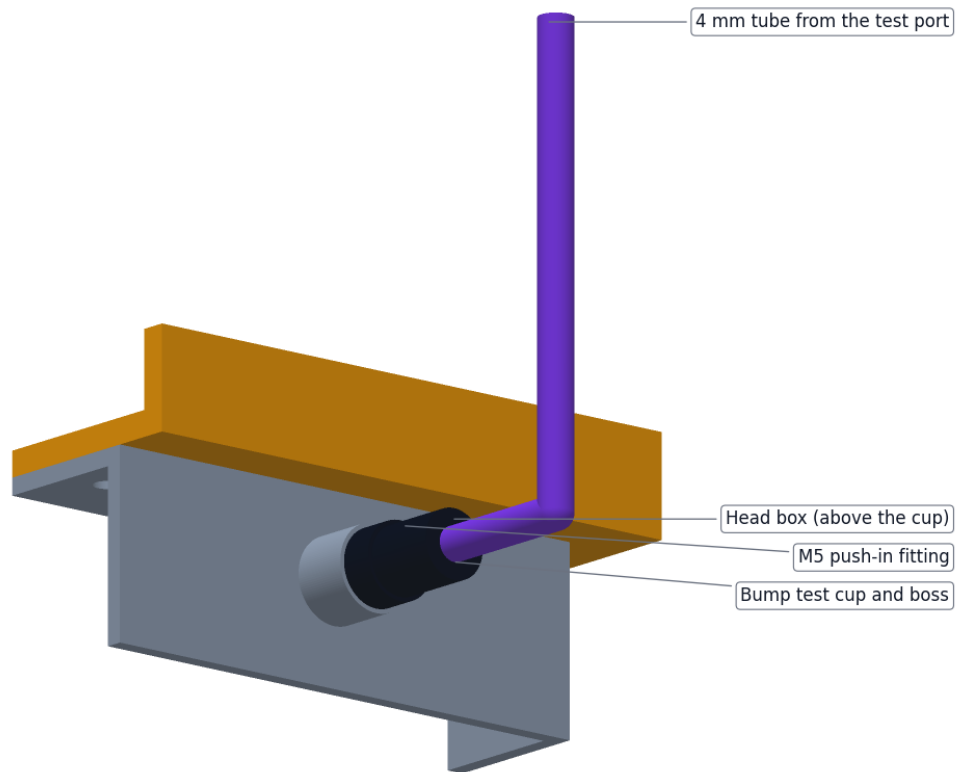
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Figure 5. Cut through a standoff. An M3 screw from inside the cup passes up through the top plate and the head floor into the standoff; the board screw goes into its other end.

The top plate sits flat against the underside of the head floor with its openings under the ports. Four M3 screws, heads inside the cup, go up through the plate and the floor into four 22 mm M3 female standoffs standing on the inside of the floor. The cup holds about 114 mL.

Joint 3: the tube at the cup

Seen from the front right and below. The tube pushes into the fitting; the fitting screws into the boss



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Figure 6. The 4 mm tube pushes into the fitting on the cup's boss.

Check before moving on. Blow gently through a length of 4 mm tube pushed into the fitting: air comes out of the cup, and the fitting does not leak round its thread.

3.3 Sensor board with both sensors

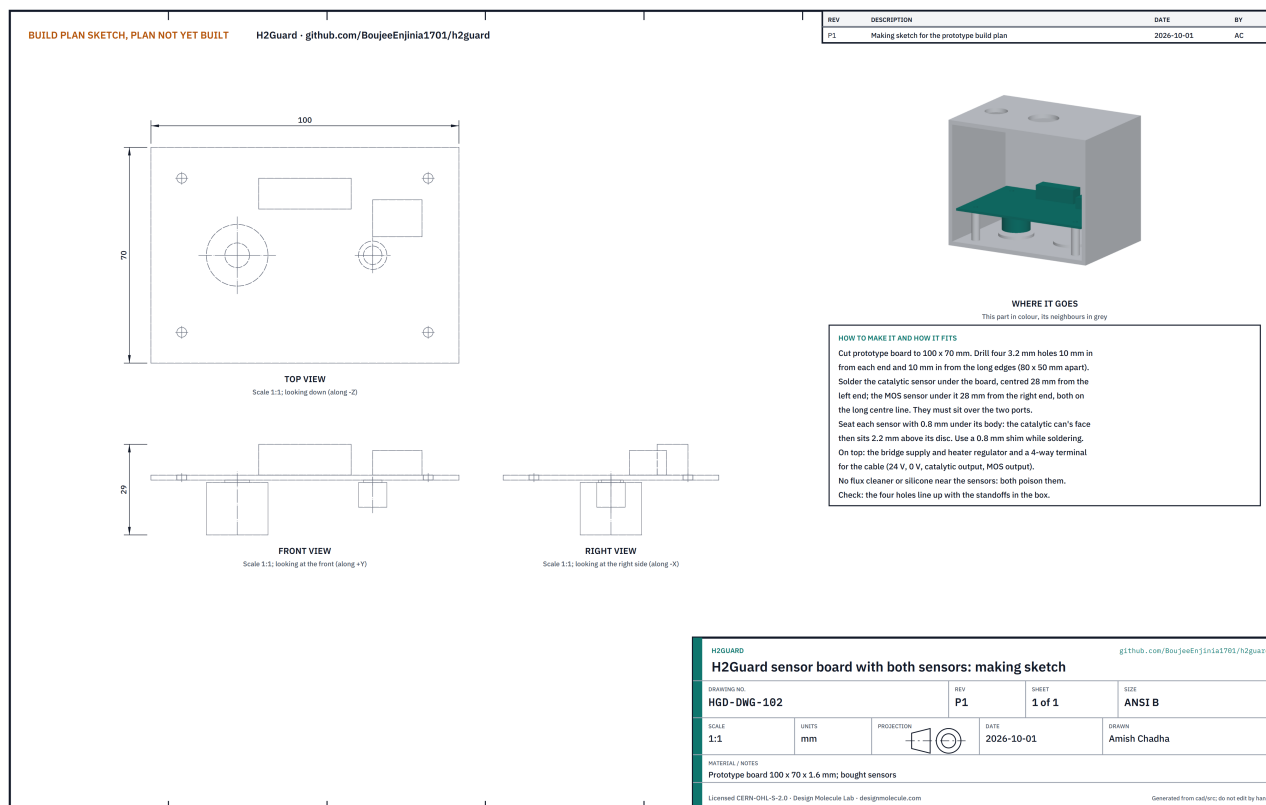


Figure 7. Sensor board making sketch (HGD-DWG-102).

What it is and what it is made from. A small board that carries the catalytic sensor and the metal oxide sensor under it, over the two ports, and their supply and output parts on top. Prototype board 100 x 70 x 1.6 mm, with the two bought sensors.

How to make it.

1. Cut prototype board to 100 x 70 mm. Drill four 3.2 mm holes 10 mm in from each end and 10 mm in from the long edges (80 x 50 mm apart), to match the standoffs.
2. Solder the catalytic sensor under the board on the long centre line, centred 28 mm from the left end; solder the metal oxide sensor under the board on the same line, 28 mm from the right end. They must sit over the left and right ports.
3. Seat each sensor 0.8 mm under the board, using a 0.8 mm shim while soldering. The catalytic sensor's face then sits 2.2 mm above its disc, which keeps the gas delay through the disc to about 1.5 s.
4. On top, build the bridge supply for the catalytic sensor, the heater supply and load resistor for the metal oxide sensor, and a 4-way screw terminal for the cable (24 V, 0 V, catalytic output, metal oxide output), following each sensor maker's application note.
5. Do not use flux cleaner, silicone or solvent near the sensors.

How it fits the parts next to it. The board stands on the four standoffs (Figure 5) and is held by four M3 screws from above. It goes in through the open front of the head box before the lid is fitted. The cable comes in through the gland in the top of the box to the 4-way terminal.

Check before moving on. With the board on its standoffs, each sensor sits centred over its port and the catalytic sensor's face is 2 to 2.5 mm above its disc (measure with feeler gauges through the open front).

3.4 Ceiling drop rod

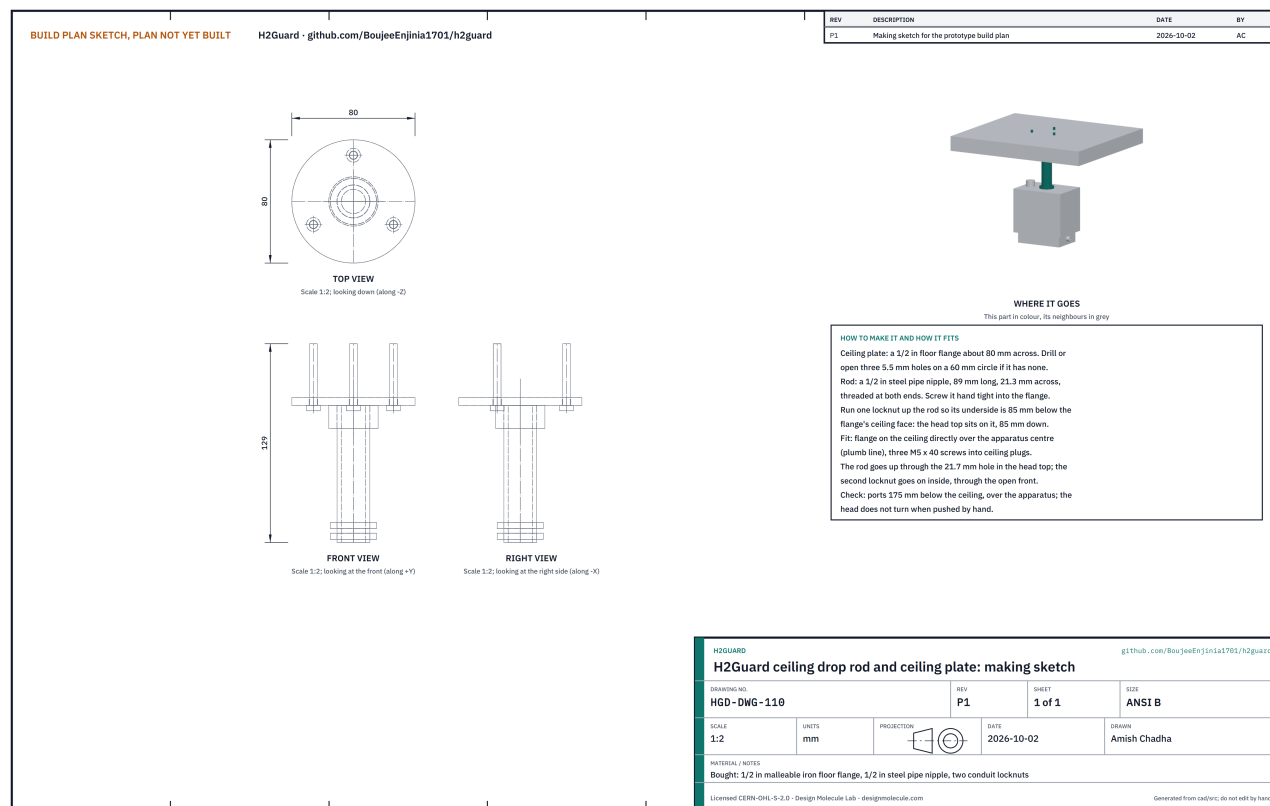


Figure 8. Ceiling drop rod and ceiling plate making sketch (HGD-DWG-110).

What it is and what it is made from. The short rod that hangs the detector head from the ceiling directly over the apparatus. A bought 1/2 in malleable iron floor flange about 80 mm across serves as the ceiling plate; a 1/2 in steel pipe nipple 89 mm long (21.3 mm across) is the rod; two 1/2 in conduit locknuts clamp the top of the head box between them.

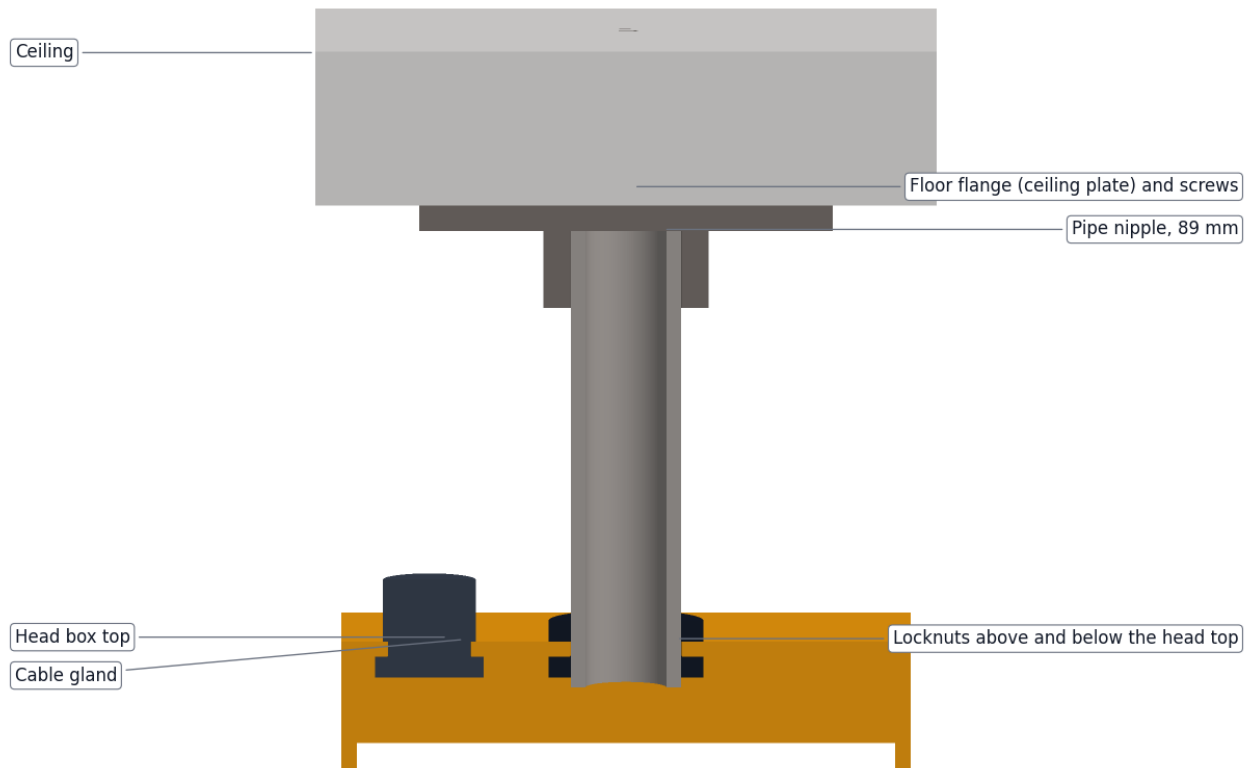
How to make it.

1. If the flange has no screw holes, drill three 5.5 mm holes on a 60 mm circle.
2. Screw the nipple hand tight into the flange, then a quarter turn with a pipe wrench.
3. Run one locknut up the nipple so its underside is 85 mm below the face of the flange that goes against the ceiling. The head's top sits on it.

How it fits the parts next to it.

Joint 8: the head on its ceiling drop rod (cut through the rod)

Seen from the front. The rod screws into the flange; two locknuts clamp the head top between them



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Figure 9. Cut through the rod. The rod screws into the flange on the ceiling; the head's top is clamped between the two locknuts.

Find the apparatus centre on the ceiling with a plumb line and fix the flange there with three M5 x 40 screws into ceiling plugs. The rod passes up through the 21.7 mm hole in the head's top; the second locknut goes on inside the box through its open front (step 4). The cable gland sits 38 mm to the left of the rod, so the cable runs along the ceiling to it.

Check before moving on. The rod hangs plumb, the flange sits flat on the ceiling, and with the head fitted its sensor ports are 175 mm below the ceiling, give or take 10 mm, and over the apparatus centre.

3.5 Controller box, drilled

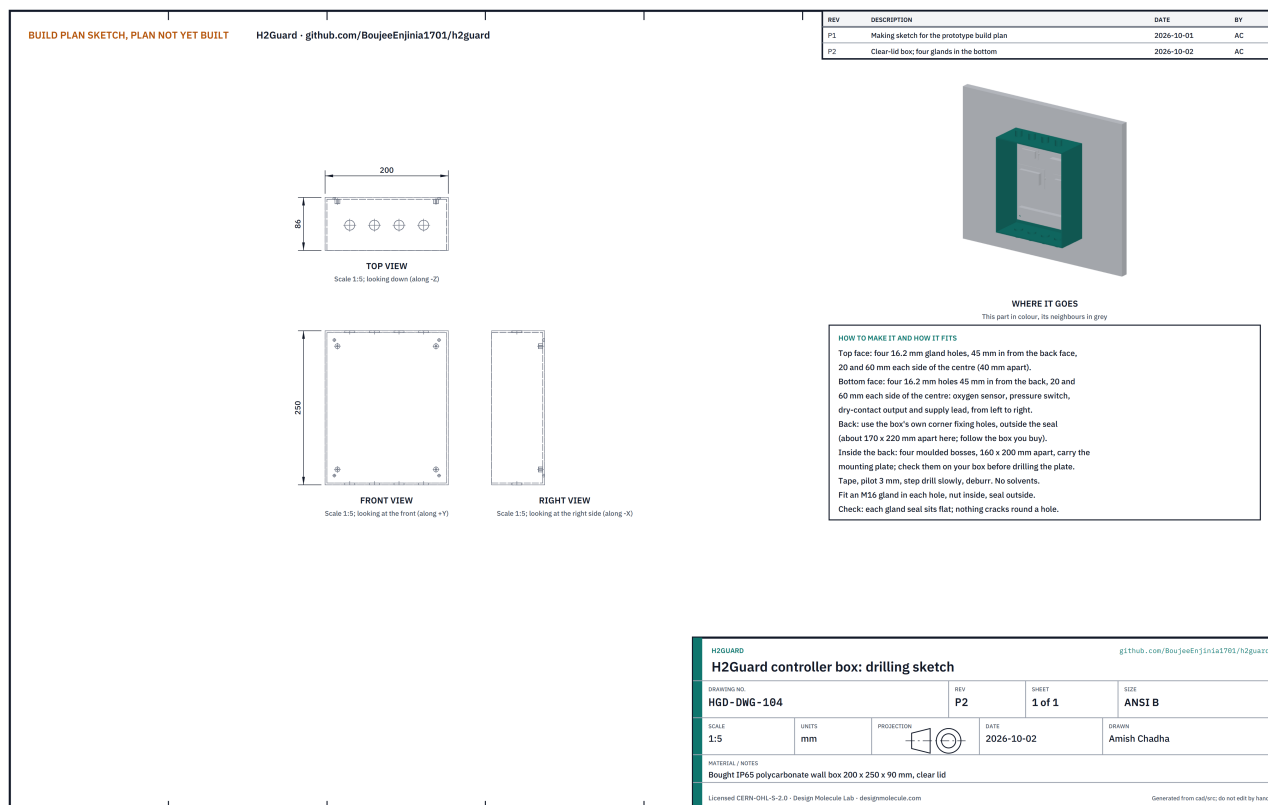


Figure 10. Controller box drilling sketch (HGD-DWG-104).

What it is and what it is made from. The wall box that holds the electronics. A bought IP65 polycarbonate box 200 wide, 250 tall and 90 deep with a clear lid on the front, four moulded bosses inside the back and fixing holes at the back corners outside the seal.

How to make it.

1. Tape the top and bottom faces and mark the holes.
2. Top: four 16.2 mm holes 45 mm in from the back face, 20 and 60 mm each side of the centre (40 mm apart), for the cables from the head, fan, valve and beacon.
3. Bottom: four 16.2 mm holes 45 mm in from the back face, 20 and 60 mm each side of the centre: from left to right, the oxygen sensor, the pressure switch, the dry-contact output and the supply lead. Leave the left one plugged if the oxygen sensor is not fitted.
4. Pilot drill 3 mm, open with a step drill slowly, deburr. No solvents: polycarbonate crazes.
5. Fit an M16 cable gland in each hole, seal outside, nut inside.
6. Measure the four bosses inside the back (the model assumes 160 mm apart across and 200 mm apart up and down) and note the result for the mounting plate.

How it fits the parts next to it. The back sits flat on the wall, top 1,475 mm above the floor (1,025 mm below the ceiling), on four screws with wall plugs through the corner fixing holes. The mounting plate sits on the bosses inside (Figure 12) and the lid closes the front (section 3.7).

Check before moving on. Every gland seal sits flat and no crack runs from any hole under a bright lamp.

3.6 Controller mounting plate and modules

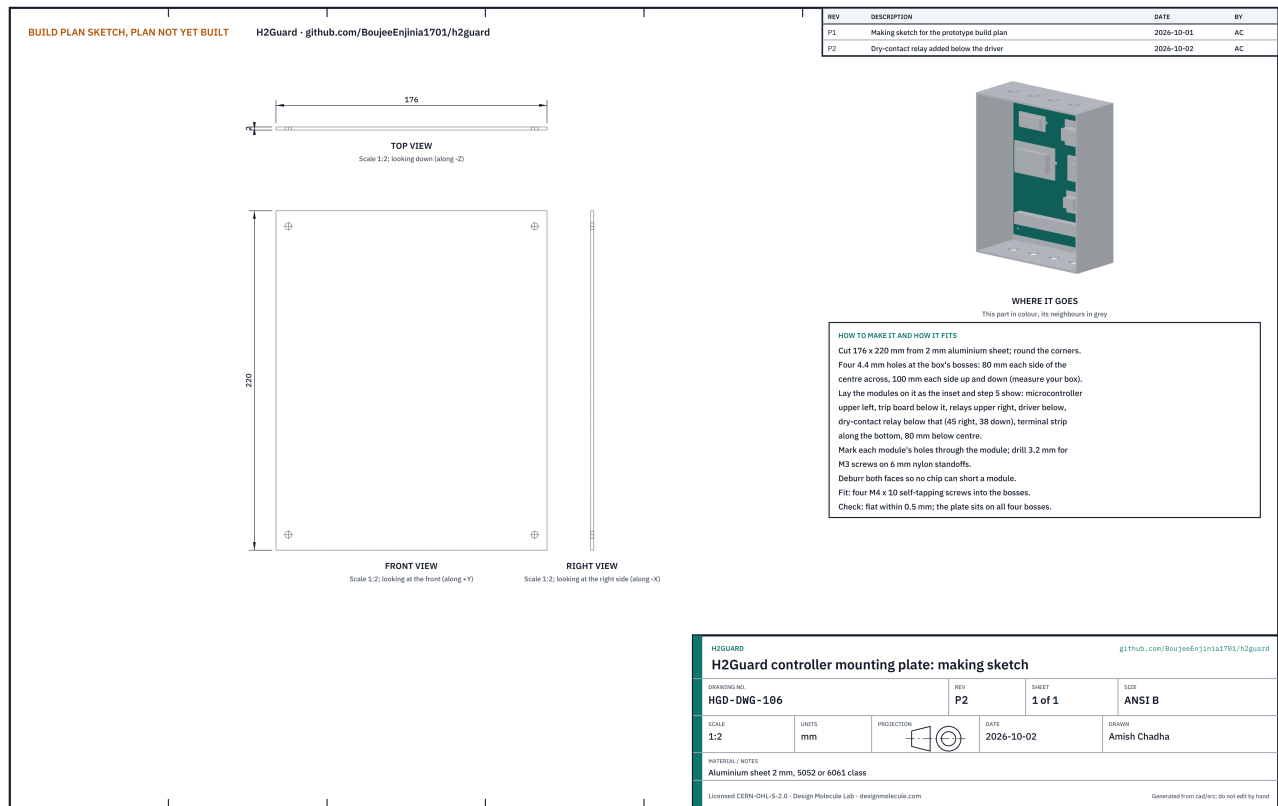


Figure 11. Controller mounting plate making sketch (HGD-DWG-106).

What it is and what it is made from. The flat plate inside the controller box that carries the bought modules, the hand-built trip board, the dry-contact relay and the terminal strip. Aluminium sheet 2 mm, 5052 or 6061 class, 176 x 220 mm.

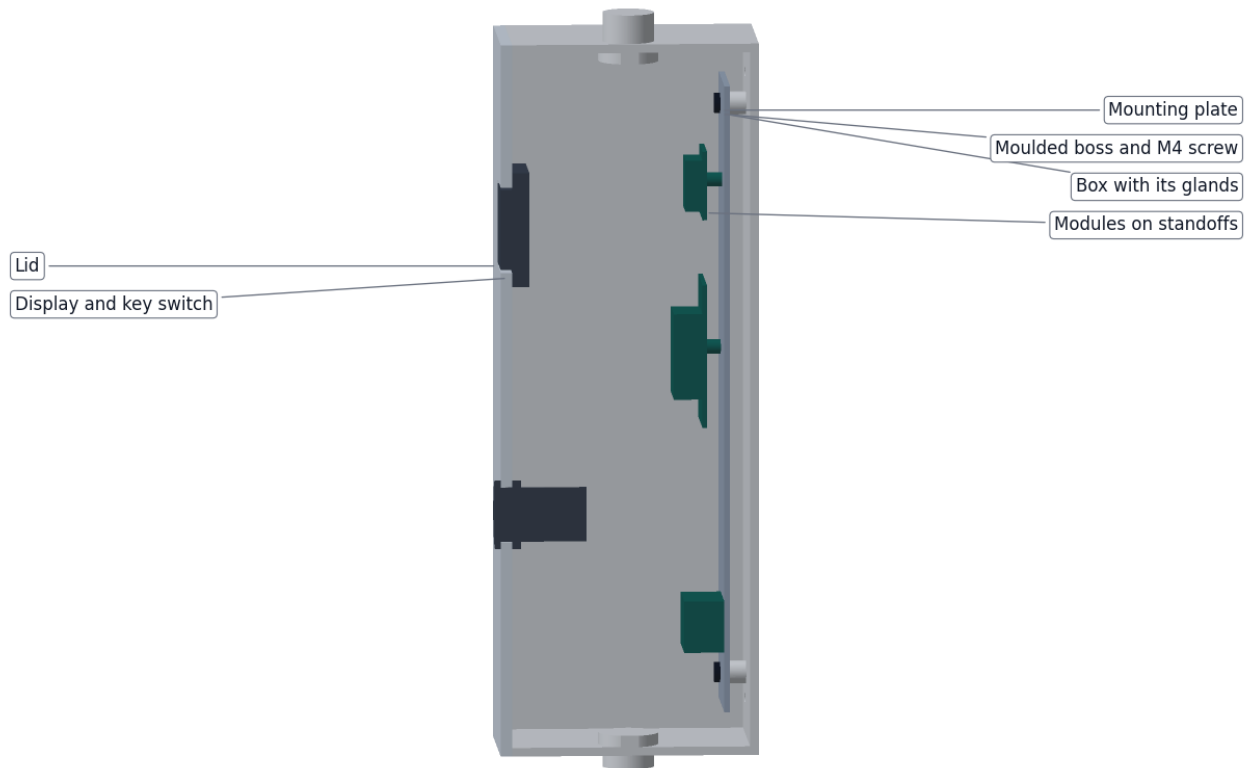
How to make it.

1. Cut the plate to 176 x 220 mm and round the corners to about 3 mm.
2. Drill four 4.4 mm holes at the boss positions you measured (the model has them 80 mm each side of the centre across and 100 mm each side up and down).
3. Lay out the modules on the front face as Figure 12 shows: microcontroller module upper left, trip board below it, relay module upper right, valve and alarm driver below it, the dry-contact relay with its 2-way output terminal below the driver (45 mm right of and 38 mm below the centre), terminal strip along the bottom, 80 mm below the centre.
4. Mark each module's mounting holes through the module, drill 3.2 mm, and fit each module on M3 screws with 6 mm nylon standoffs. Fit the terminal strip directly to the plate.
5. Deburr both faces so no chip can short a module.
6. Wire the modules as Figure 13 shows (section 3.6.1).

How it fits the parts next to it.

Joint 4: controller cut open through the key switch

Seen from the right. Plate on the bosses at the back, modules on standoffs, panel parts in the lid at the front



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Figure 12. The plate sits on the four moulded bosses at the back of the box; the modules stand 6 mm off it; the display and key switch are in the lid at the front, well clear of the modules.

The plate lies flat on the four bosses, held by four M4 x 10 self-tapping screws. Its front face is 9 mm in front of the inside of the back wall, and the deepest module stands about 23 mm in front of the plate, leaving about 25 mm to the back of the key switch in the lid.

Check before moving on. The plate is flat within 0.5 mm and sits on all four bosses without rocking.

3.6.1 Wiring

H2Guard prototype: block-level wiring

Bought modules and a hand-wired trip board; no circuit board is laid out. Stranded copper; ferrules on every screw terminal. All circuits 24 V DC or lower.

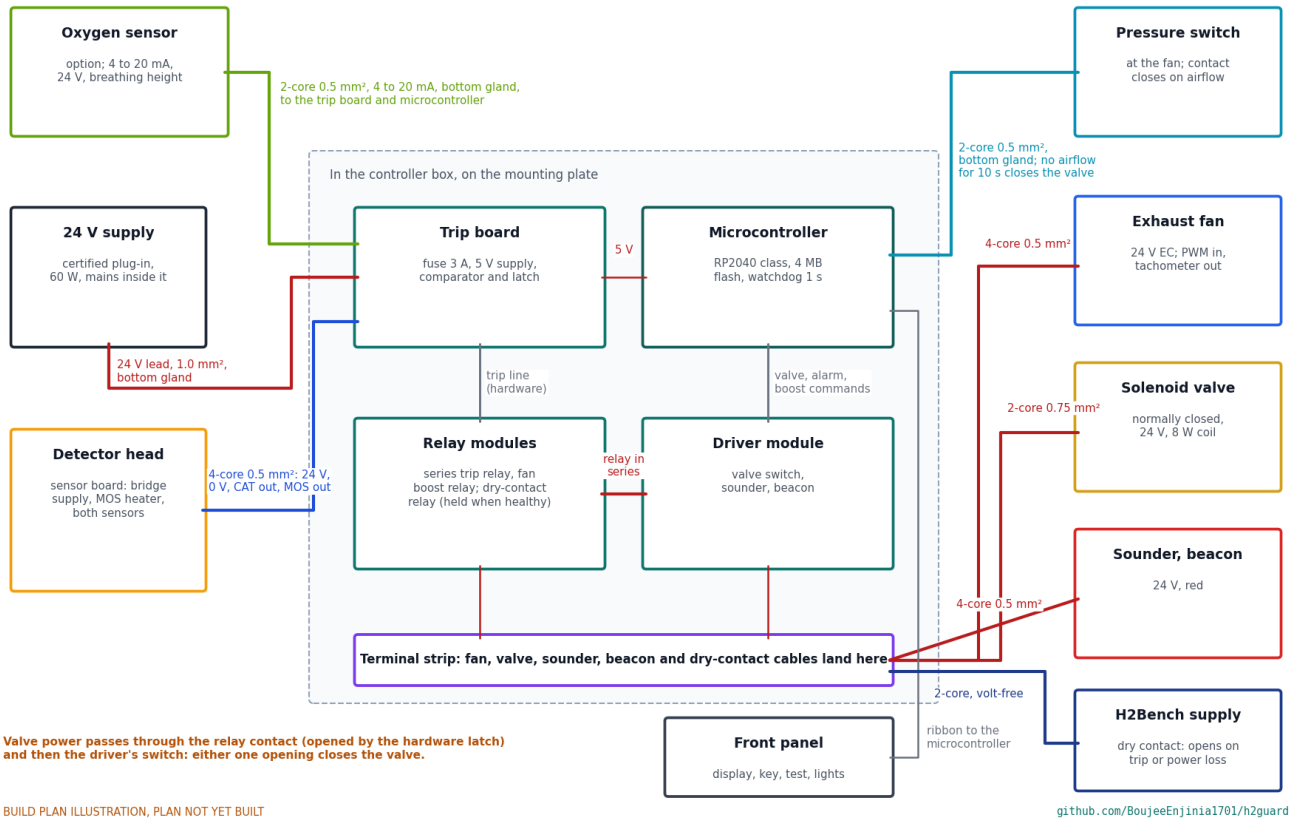


Figure 13. Block-level wiring with wire sizes. No circuit board is laid out at this stage; bought modules and a hand-wired trip board stand in for the controller board.

The controller board in the bill of materials is a custom board, which is TRL 4 work. For this prototype, buy or build these:

Table 2. Modules that stand in for the controller board.

Module	What to buy or build
Microcontroller	RP2040-class module with 4 MB of flash or more and a hardware watchdog
Trip board	Hand-wired on prototype board: input fuse about 3 A, 5 V supply from 24 V, the catalytic sensor input, a window comparator that trips above the 25 % LFL level or when the input reads open or shorted, a latch that only the key resets, and a 150 ohm resistor that turns the oxygen sensor's 4 to 20 mA signal into a voltage for the microcontroller
Relay module	Two relays with 24 V coils: the series trip relay, opened by the latch, and the fan boost relay
Dry-contact relay	One relay with a 24 V coil and a changeover contact rated 2 A at 30 V DC, held energized while all is well, with a 2-way screw terminal for the output
Driver module	Low-side switches for the valve coil (with a Zener clamp for fast closing), the sounder and the beacon, driven by the microcontroller
Terminal strip	5.08 mm pitch screw terminals for every field cable

Wire it like this, with stranded copper and a ferrule on every screw terminal:

1. Supply lead from the 24 V unit through the bottom gland to the trip board's fused input: 1.0 mm² (17 AWG).
2. Trip board 5 V output to the microcontroller module: 0.5 mm² (20 AWG).
3. Head cable, 4-core 0.5 mm², through a top gland to the trip board: 24 V, 0 V, catalytic output, metal oxide output.
4. Valve coil supply from the terminal strip through the series trip relay contact and then the driver's valve switch: 0.75 mm² (18 AWG). Either one opening closes the valve.
5. Fan cable, 4-core 0.5 mm², from the terminal strip: 24 V, 0 V, speed signal and tachometer.
6. Sounder and beacon cable, 4-core 0.5 mm², from the terminal strip.
7. Front panel parts to the microcontroller with a short ribbon cable, 0.14 mm².
8. Pressure switch cable, 2-core 0.5 mm², through a bottom gland to a microcontroller input. With the fan running, if the switch has not closed for 10 s, the microcontroller closes the valve, latches and shows the fault.
9. Dry-contact output cable, 2-core 0.5 mm², from the dry-contact relay's terminal through a bottom gland to the host bench. The relay is held energized while all is well, so its contact opens on a trip, a fault or loss of power.
10. Oxygen sensor cable (if fitted), 2-core 0.5 mm², through a bottom gland to the trip board's 4 to 20 mA input. Below 19.5 % oxygen the controller warns; below 18 % it closes the valve and latches.

Check before moving on. Every wire continues end to end; with the supply unplugged, no 24 V terminal reads short to 0 V; every wire is labelled at both ends.

3.7 Controller lid with the front panel

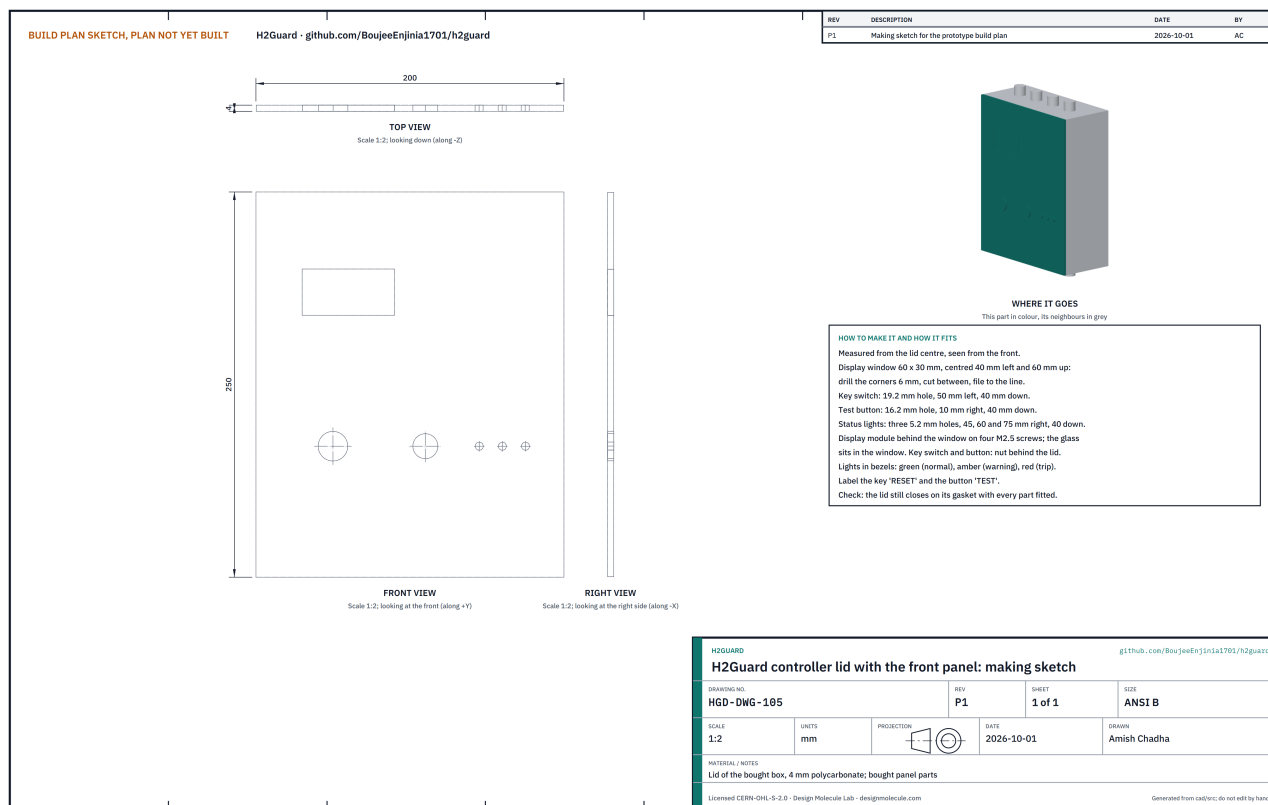


Figure 14. Controller lid making sketch (HGD-DWG-105).

What it is and what it is made from. The box's own clear 4 mm polycarbonate lid, cut to take the display, the key-switch reset, the test button and three status lights.

How to make it. All positions are from the lid centre, seen from the front.

1. Display window 60 x 30 mm, centred 40 mm left and 60 mm up: drill a 6 mm hole in each corner, cut between them and file to the line.
2. Key switch: 19.2 mm hole, 50 mm left and 40 mm down.
3. Test button: 16.2 mm hole, 10 mm right and 40 mm down.
4. Status lights: three 5.2 mm holes, 45, 60 and 75 mm right and 40 mm down.
5. Fit the display module behind the window on four M2.5 screws, the glass in the window. Fit the key switch and button through their holes with their nuts behind the lid, and the lights in bezels: green (normal), amber (warning), red (trip).
6. Label the key RESET and the button TEST.

How it fits the parts next to it. The lid closes on the box's gasket with its own captive screws; the panel parts reach about 26 mm into the box, clear of the modules (Figure 12).

Check before moving on. The lid closes evenly on its gasket with every part fitted.

3.8 Bump test port bracket

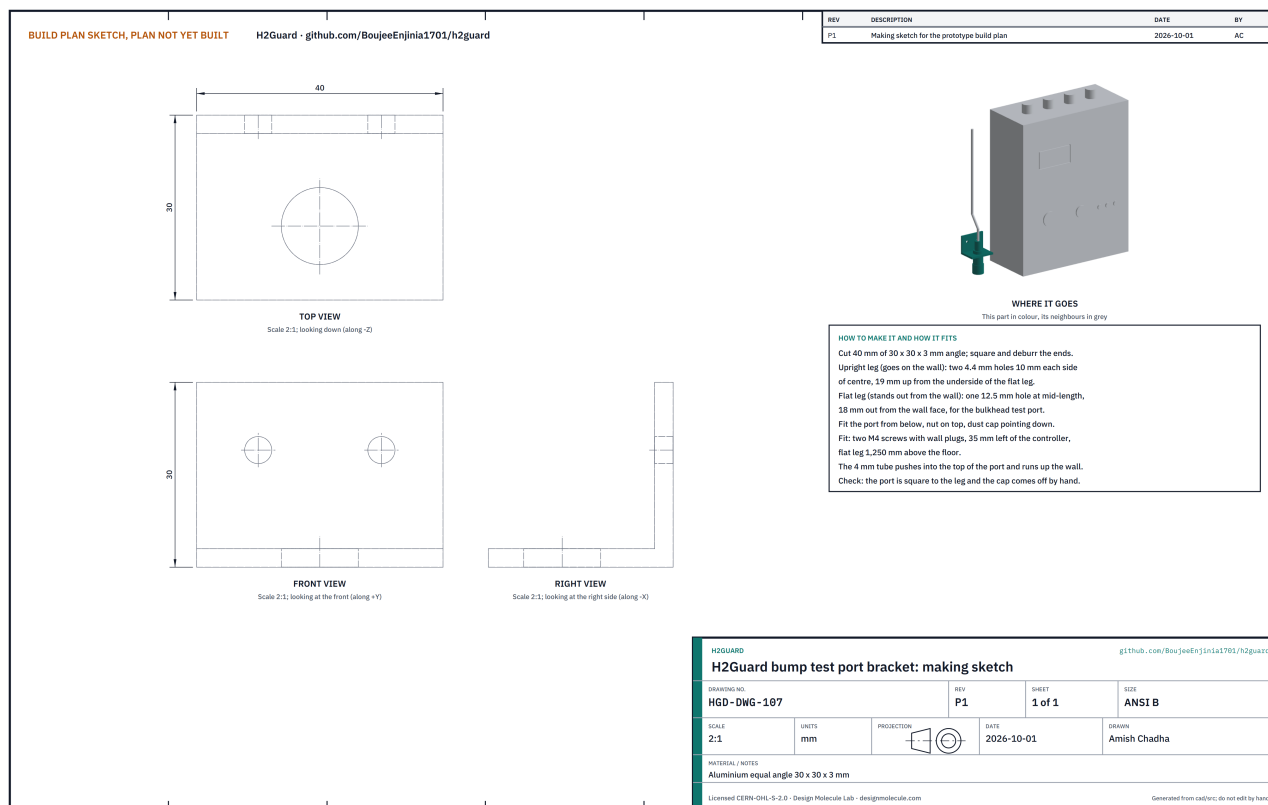


Figure 15. Test port bracket making sketch (HGD-DWG-107).

What it is and what it is made from. A short angle that holds the capped test port on the wall beside the controller. Aluminium equal angle 30 x 30 x 3 mm, 40 mm long.

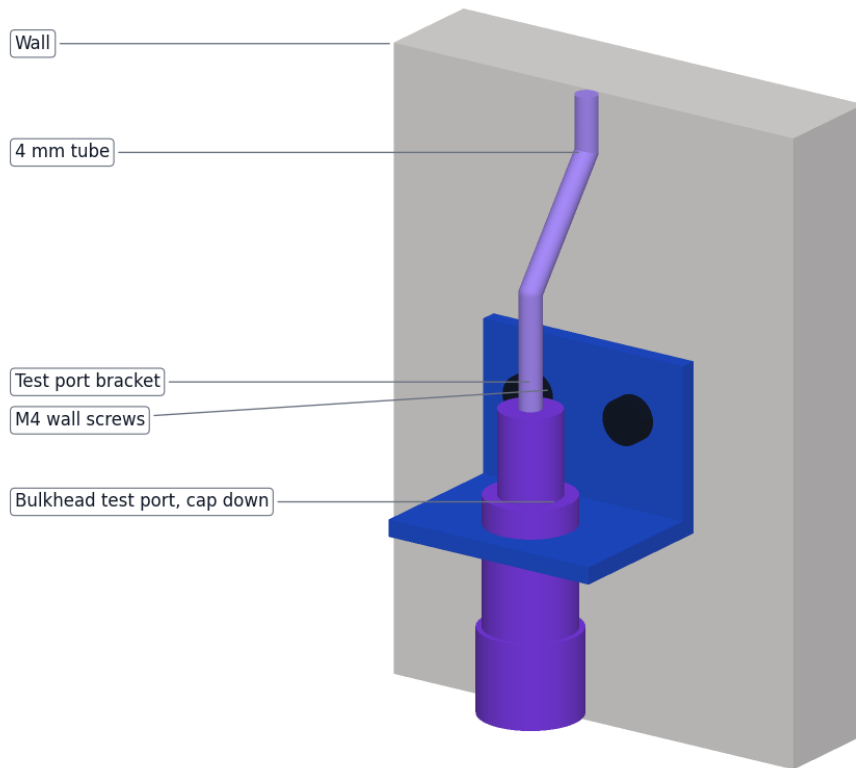
How to make it.

1. Cut 40 mm of angle; square and deburr the ends.
2. Upright leg (goes on the wall): two 4.4 mm holes 10 mm each side of the centre, 19 mm up from the underside of the flat leg.
3. Flat leg (stands out from the wall): one 12.5 mm hole at mid-length, 18 mm out from the wall face.
4. Fit the bulkhead test port up through the flat leg, nut on top, dust cap underneath.

How it fits the parts next to it.

Joint 5: bump test port on its bracket

Seen from the front left. The port goes up through the flat leg; the tube pushes into its top



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Figure 16. The port goes up through the flat leg; the tube pushes into its top.

The upright leg sits flat on the wall 35 mm left of the controller, with the flat leg 1,250 mm above the floor, on two M4 screws with wall plugs. Test gas connects to the port from below; the 4 mm tube leaves its top and runs up the wall.

Check before moving on. The port is square to the leg and the dust cap comes off by hand.

3.9 Fan plate

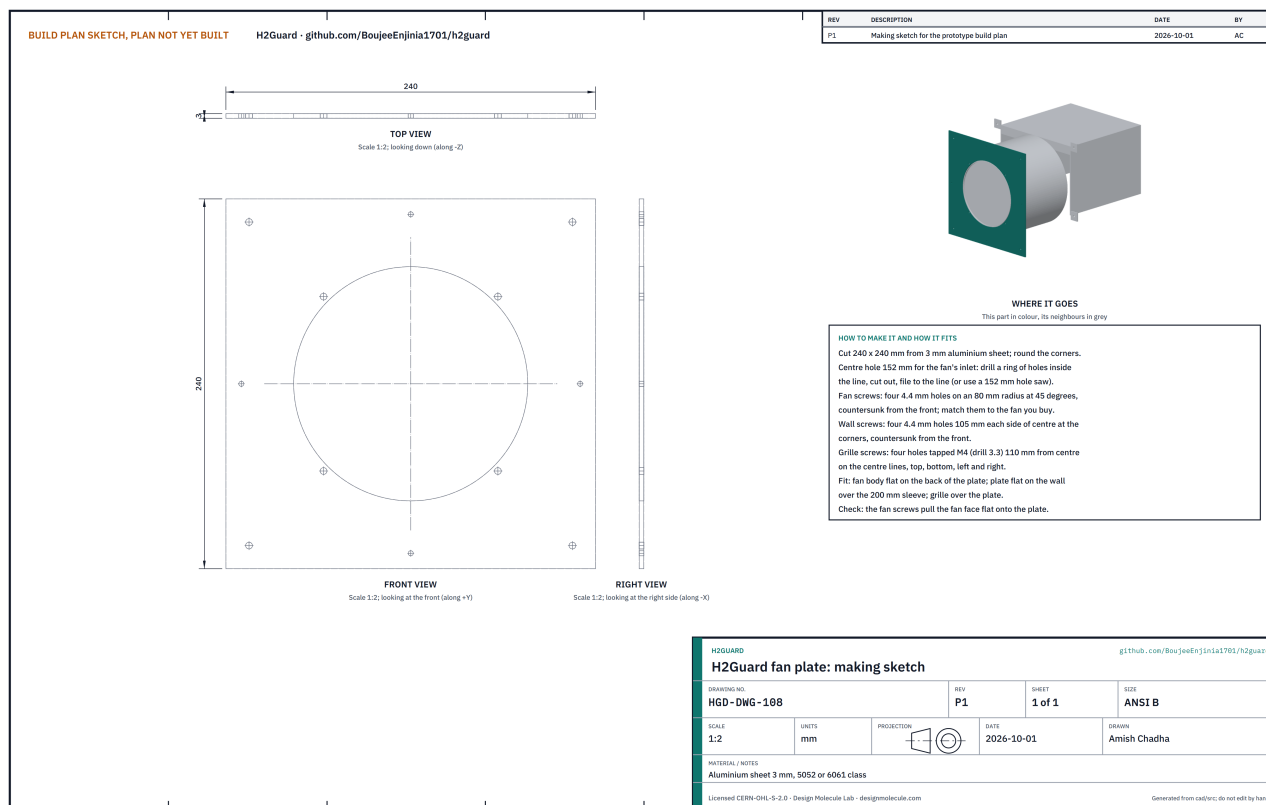


Figure 17. Fan plate making sketch (HGD-DWG-108).

What it is and what it is made from. A square plate on the inside wall face that holds the exhaust fan in its wall sleeve and carries the inside grille. Aluminium sheet 3 mm, 5052 or 6061 class, 240 x 240 mm.

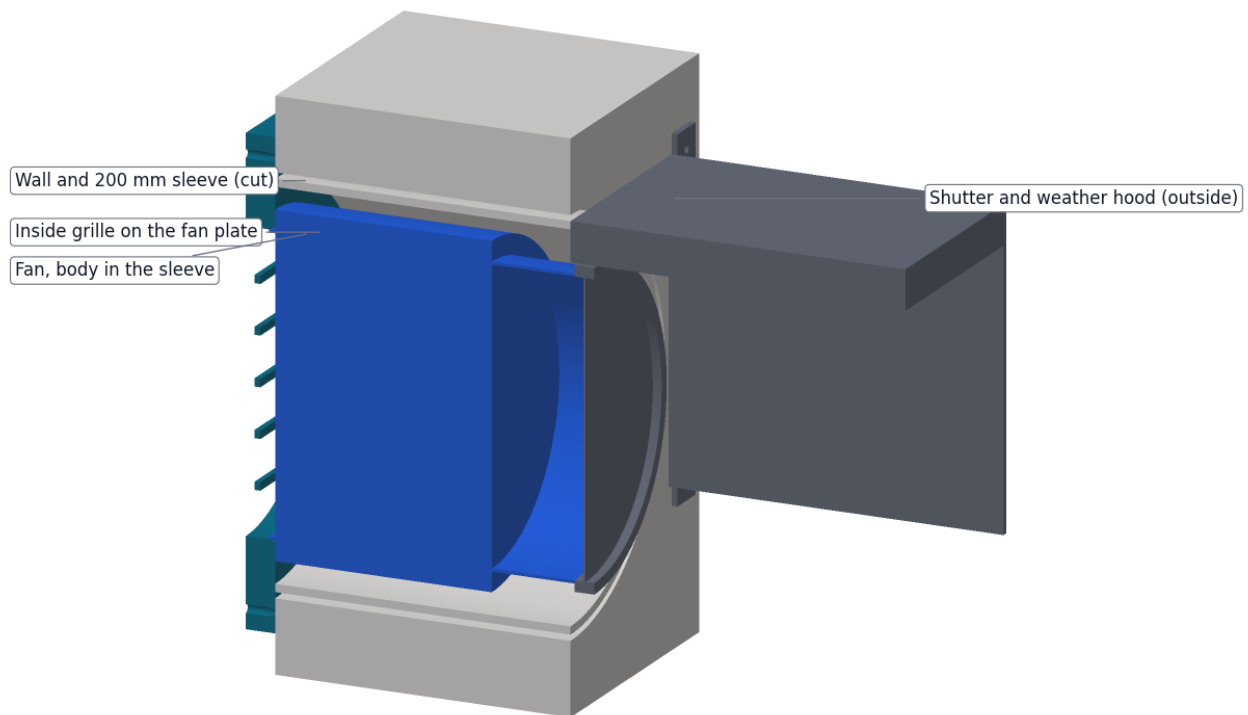
How to make it.

1. Cut the plate to 240 x 240 mm and round the corners.
2. Cut a 152 mm centre hole for the fan's inlet: drill a ring of holes just inside the line, cut out and file to the line, or use a 152 mm hole saw.
3. Fan screws: four 4.4 mm holes on an 80 mm radius at 45° to the centre lines, countersunk from the front. Match them to the fixing holes in the inlet face of the fan you buy.
4. Wall screws: four 4.4 mm holes 105 mm each side of the centre (near the corners), countersunk from the front.
5. Grille screws: four holes tapped M4 (drill 3.3 mm), 110 mm from the centre on the centre lines (top, bottom, left and right).

How it fits the parts next to it.

Joint 6: exhaust fan in the wall (cut through the fan's axis)

Seen from the right and a little from outside; the room is on the left. The fan plate holds the fan



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Figure 18. Cut through the fan's axis. The fan body sits in the 200 mm sleeve, its inlet face flat on the back of the plate; the grille covers the plate; the shutter and hood are outside.

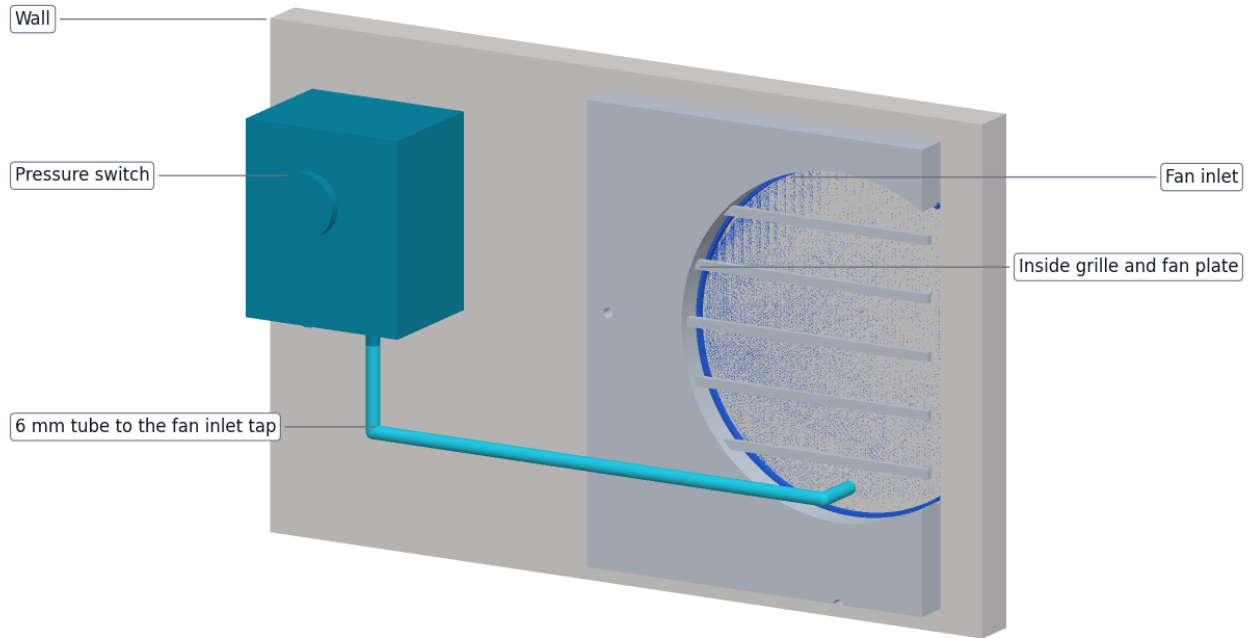
The fan's inlet face sits flat on the back of the plate, held by four M4 countersunk screws from the front. The plate sits flat on the inside wall face over the 200 mm sleeve, centred 2,300 mm above the floor, on four countersunk screws with wall plugs. The inside grille screws to the plate's tapped holes. The fan body stands 11 mm clear of the sleeve all round.

Check before moving on. The four fan screws pull the fan face flat onto the plate, and the fan turns freely by hand.

The pressure switch beside the fan. A bought low-range air pressure switch, about 85 x 85 x 55 mm with its two ports underneath, goes on the wall to the left of the fan plate, its centre 230 mm left of the fan's centre and 30 mm above it, on two screws with plugs. A 6 mm tube from its low port runs down 37 mm, across in front of the grille and into the grille opening, 62 mm below the fan's centre, ending just inside the grille in the fan's inlet. The high port stays open to the room. Air drawn into the fan lowers the pressure at the tube's end by about 6 Pa at the continuous flow and 30 Pa on boost; with no flow, whether the fan has stopped, the duct is blocked or the shutter is stuck, the difference falls to about zero. Set the switch to close at about 3 Pa.

Joint 9: pressure switch and its tap in the fan inlet

Seen from the front left. The low port's tube ends just inside the grille; the high port is open to the room



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Figure 19. The tube from the switch's low port ends just inside the grille; the other port is open to the room.

3.10 Valve bracket

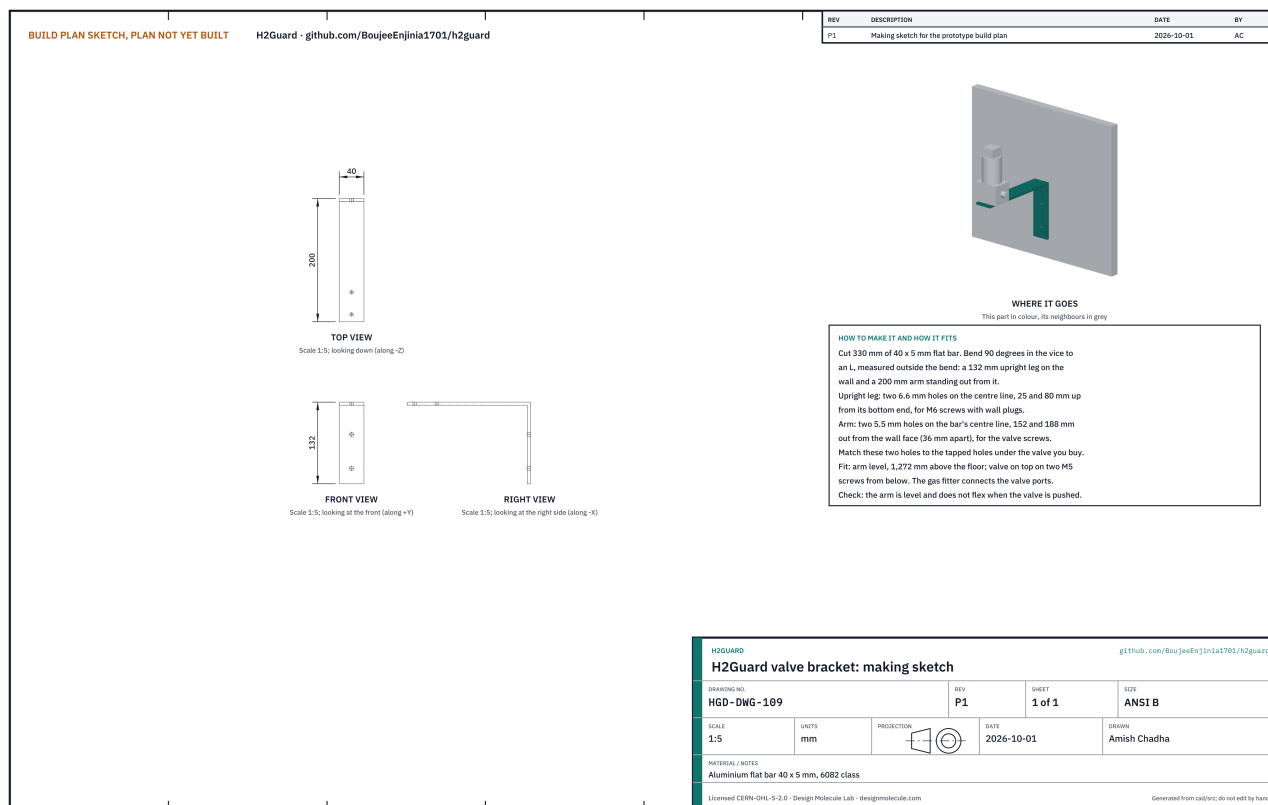


Figure 20. Valve bracket making sketch (HGD-DWG-109).

What it is and what it is made from. An L-shaped bracket that holds the solenoid valve off the wall, in line with the supply. Aluminium flat bar 40 x 5 mm, 6082 class.

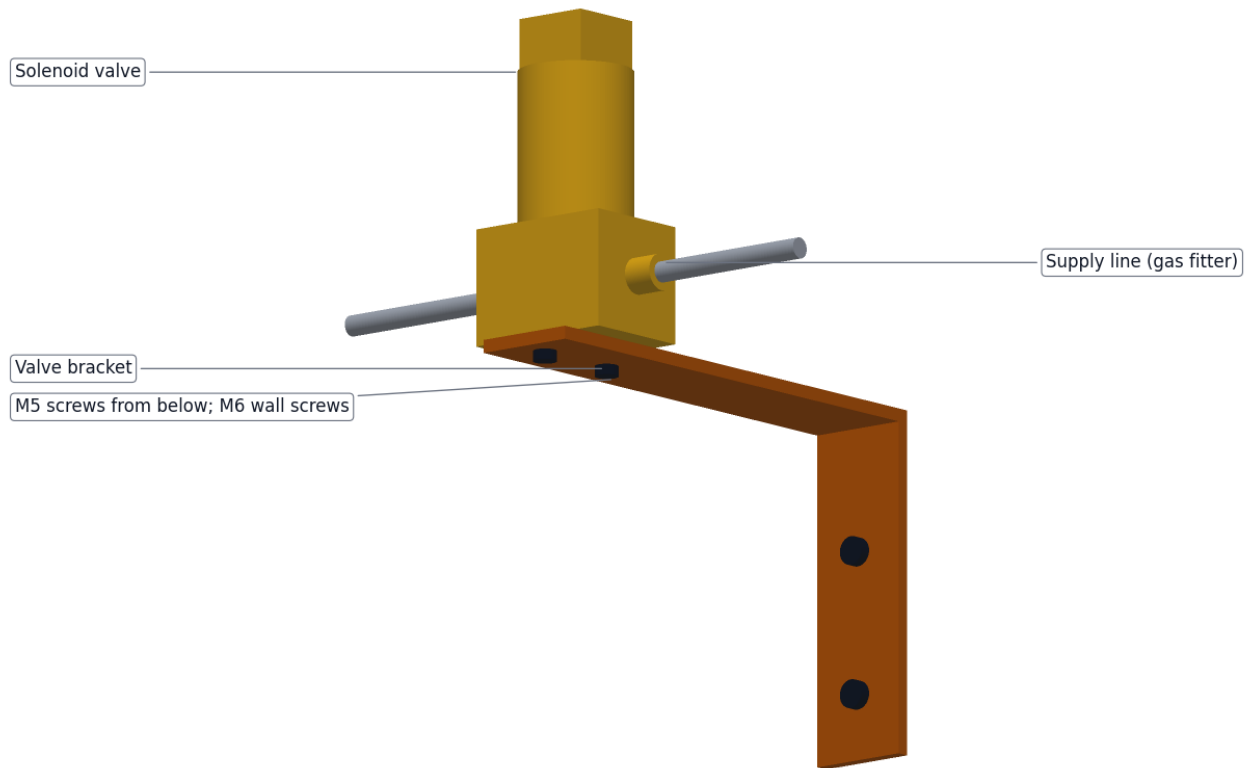
How to make it.

1. Cut 330 mm of bar. Bend it 90° in the vice to an L, measured outside the bend: a 132 mm leg for the wall and a 200 mm arm. Bend slowly round a radiused jaw so the outside of the bend does not crack.
2. Wall leg: two 6.6 mm holes on the centre line, 25 and 80 mm up from its bottom end.
3. Arm: two 5.5 mm holes on the centre line, 152 and 188 mm out from the wall face (36 mm apart). Match them to the tapped holes under the valve you buy.

How it fits the parts next to it.

Joint 7: valve on its bracket

Seen from the front left and below. Two screws up through the arm into the valve body



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Figure 21. Two M5 screws come up through the arm into the valve body; the gas fitter connects the supply to the valve's ports.

The wall leg sits flat on the wall on two M6 screws with plugs; the arm is level, its top 1,277 mm above the floor. The valve sits on the arm, held by two M5 screws from below. The supply line connects to the valve's two ports, downstream of the regulator and flow restrictor.

Check before moving on. The arm is level and does not flex when you press on the valve.

3.11 Bought components

Buy to specification, not brand. Line numbers are those of the bill of materials.

- **Head box (line 1).** ABS or polycarbonate, 110 x 80 x 90 mm, lid on a 110 x 90 face, with an M16 cable gland. Drill as section 3.1.
- **Ceiling drop rod (line 17).** A 1/2 in malleable iron floor flange, a 1/2 in steel pipe nipple 89 mm long, two 1/2 in conduit locknuts, three M5 x 40 screws with ceiling plugs. Make up as section 3.4.
- **Catalytic and metal oxide sensors (lines 2 and 3).** A catalytic sensor reading 0 to 100 % LFL of hydrogen with a can about 20 mm across, and a metal oxide hydrogen sensor for about 30 to 3,000 ppm, each with its maker's datasheet and application note.

- **Arrestor discs (line 4).** Two sintered stainless steel discs 25 mm across and 2 mm thick, pores 50 µm or finer; two-part epoxy.
- **Controller box (line 5).** IP65 polycarbonate, 200 x 250 x 90 mm, clear lid on the front, four moulded bosses inside the back, corner fixing holes outside the seal; eight M16 cable glands.
- **Controller modules (line 6) and front panel parts (line 7).** As Table 2; a small display module, a key switch for a 19 mm hole, a push button for a 16 mm hole and three 5 mm lights in bezels.
- **24 V supply (line 8).** Certified plug-in or desktop unit, 24 V, 2.5 A (60 W).
- **Exhaust fan (line 9).** 150 mm mixed-flow EC duct fan, 24 V, about 450 m³/h in free air and 200 Pa at shut-off, speed input and tachometer output, body about 170 mm across with fixing holes in its inlet face; a 240 mm square inside grille; a 200 mm wall sleeve 150 mm long; a backdraft shutter for the 150 mm outlet; a weather hood with fixing tabs.
- **Make-up air grille (line 10).** Wall grille 300 x 160 mm with insect mesh and about 50 % free area.
- **Solenoid valve (line 11).** Two-way, normally closed, direct acting, 1/4 in, stainless steel body with FKM or PTFE seals and two tapped mounting holes underneath, 24 V DC coil of about 8 W, 0 to 10 bar. Its maker must state that it is suitable for hydrogen and give a seat leak rate. A low-cost brass valve may stand in only if its maker states both and it passes a seat leak proof test.
- **Pressure switch (line 18).** A low-range adjustable air pressure switch that can be set to about 3 Pa, with a changeover contact for 24 V DC and two barbs for 6 mm tube; about 0.5 m of 6 mm tube.
- **Dry-contact relay (line 19).** A relay module with a 24 V coil and a changeover contact rated 2 A at 30 V DC, and a 2-way screw terminal.
- **Oxygen sensor (line 20, only where inert gas cylinders share the room).** An electrochemical oxygen transmitter, 0 to 25 %, 24 V DC, 4 to 20 mA output, in an IP65 box with the cell facing down; 5 m of two-core cable.
- **Fan for a room used without close supervision (line 21).** In such a room, fit a 150 mm duct fan with its motor outside the air stream (a bifurcated fan) in place of line 9, of the same delivery. If it runs on mains, an electrician wires it through a certified contactor switched by the controller's fan relay.
- **Sounder and beacon (line 12).** 24 V DC, about 90 dB at 1 m, red flashing beacon.
- **Cable (line 13).** Four-core 0.5 mm² for the head, fan and beacon; two-core 0.75 mm² for the valve; two-core 0.5 mm² for the pressure switch and the dry-contact output; about 22 m in all, with surface conduit.
- **Fixings (line 14).** Four M3 x 22 mm female standoffs; M3 screws; 6 mm nylon standoffs; M2.5, M4, M5 and M6 screws; wall plugs; ferrules, heat-shrink and labels.
- **Test port, tube and fittings (line 15).** Bulkhead push-fit test port for 4 mm tube with a 12 mm thread and dust cap; about 3.5 m of 4 mm PTFE or nylon tube; an M5 straight push-in fitting; six saddle clips for the wall and two for the ceiling.
- **Bar and sheet (line 16).** 2 mm and 3 mm aluminium sheet, 30 x 30 x 3 mm angle and 40 x 5 mm flat bar for the made parts.

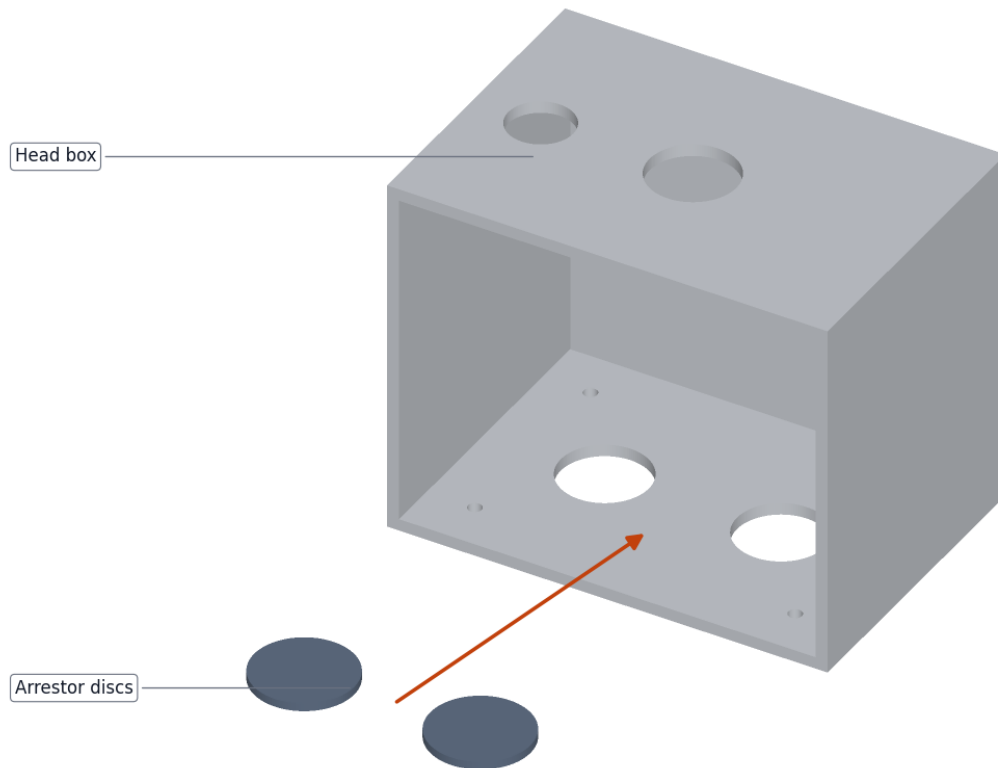
■ 4. Putting it together

In each picture the parts already fitted are grey and the part being fitted is in colour, with an arrow showing the way it goes in. Steps 1 to 3, 5, 6 and 8 are bench work; the rest is done in the room. Step 15 applies only where inert gas cylinders share the room.

Step 1: arrestor discs into the head

Step 1: arrestor discs into the head

Through the open front; each disc on its port ledge with a bead of two-part epoxy round its edge. No silicone



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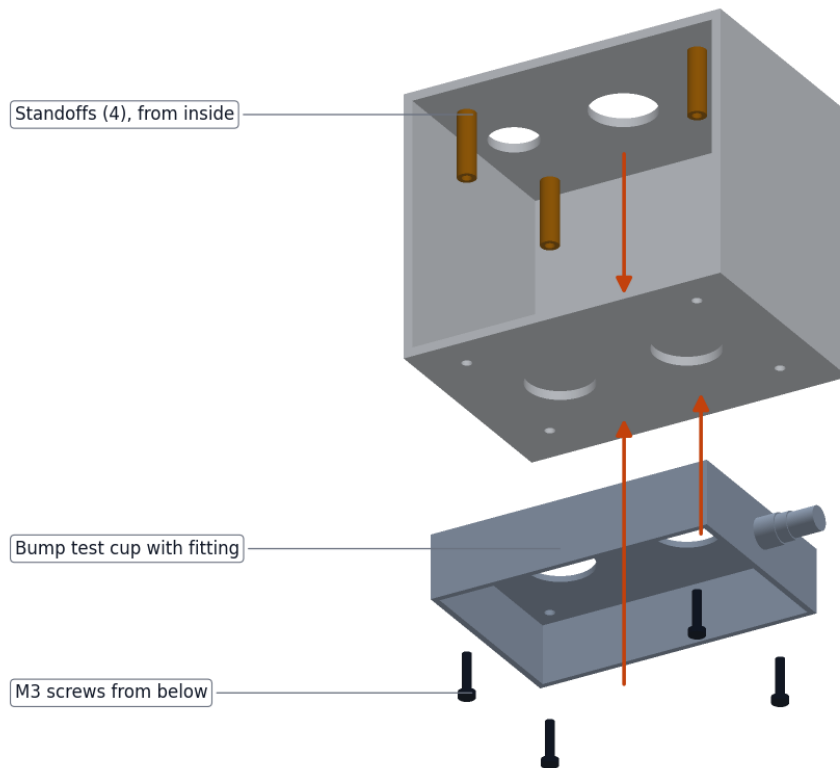
github.com/BoujeeEnjinia1701/h2guard

Through the open front, lay each disc on its port ledge and run a bead of two-part epoxy round its edge. Let it cure fully, with the box open to the air, before the sensors go in.

Step 2: standoffs and bump test cup

Step 2: standoffs and bump test cup

Hold each standoff inside; screw up through the cup and the floor into it. Seen from below



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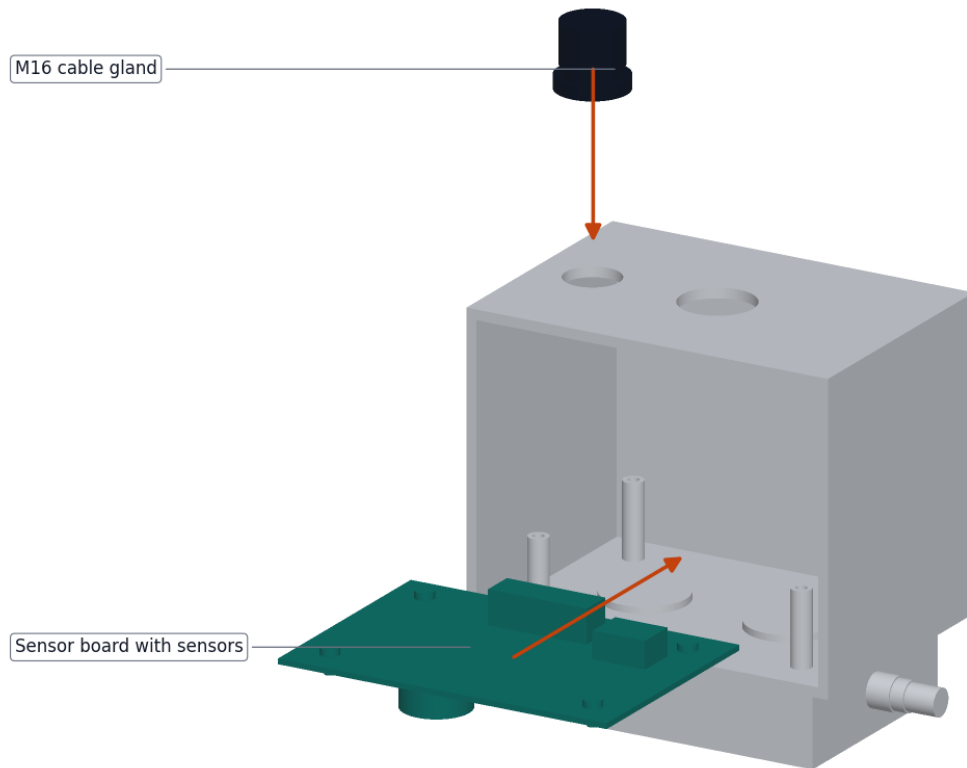
github.com/BoujeeEnjinia1701/h2guard

Hold each standoff on the inside of the floor and screw an M3 screw up through the cup's top plate and the floor into it. Check the cup's openings sit under the ports before tightening.

Step 3: sensor board and cable gland

Step 3: sensor board and cable gland

Board in through the open front onto the standoffs, four M3 screws; gland in the top, nut inside



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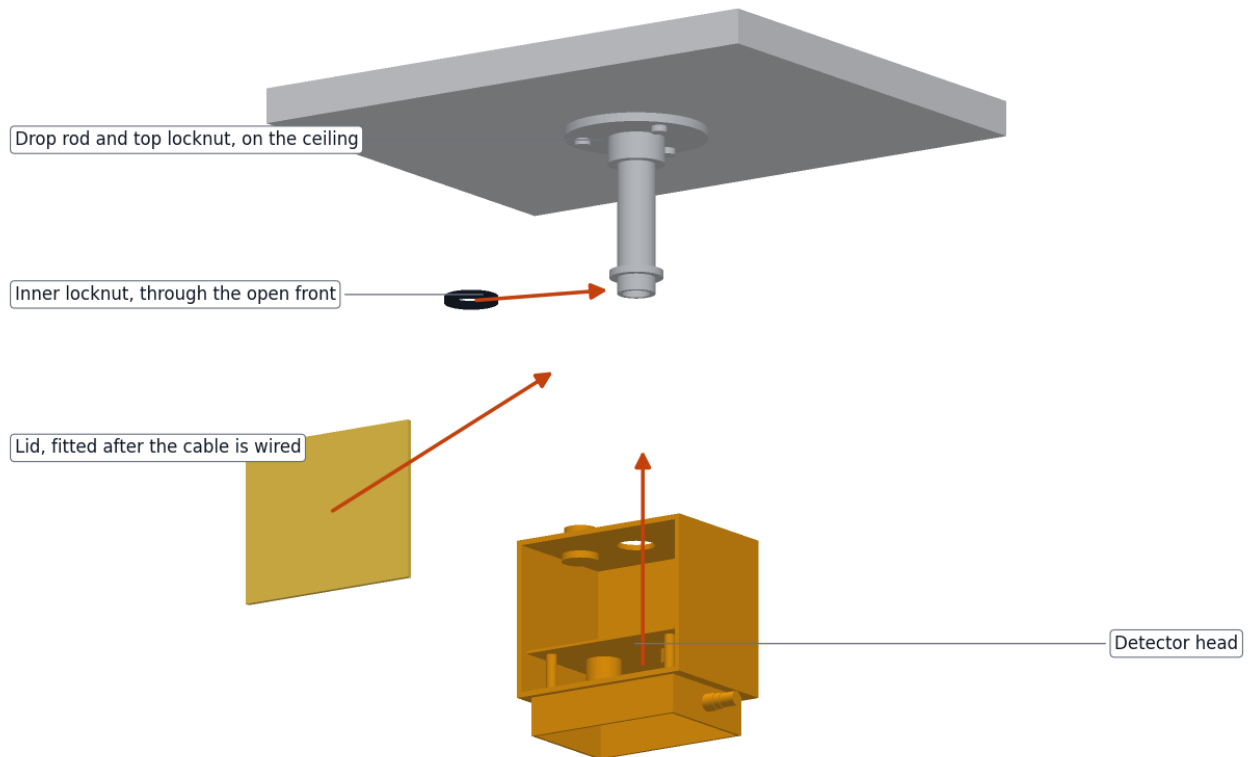
github.com/BoujeeEnjinia1701/h2guard

Slide the board in through the open front onto the standoffs and fit four M3 screws. Fit the cable gland in the top, nut inside.

Step 4: detector head onto the ceiling drop rod

Step 4: detector head onto the ceiling drop rod

Seen from below. Flange screwed to the ceiling over the apparatus first; push the head up onto the rod and fit the inner locknut



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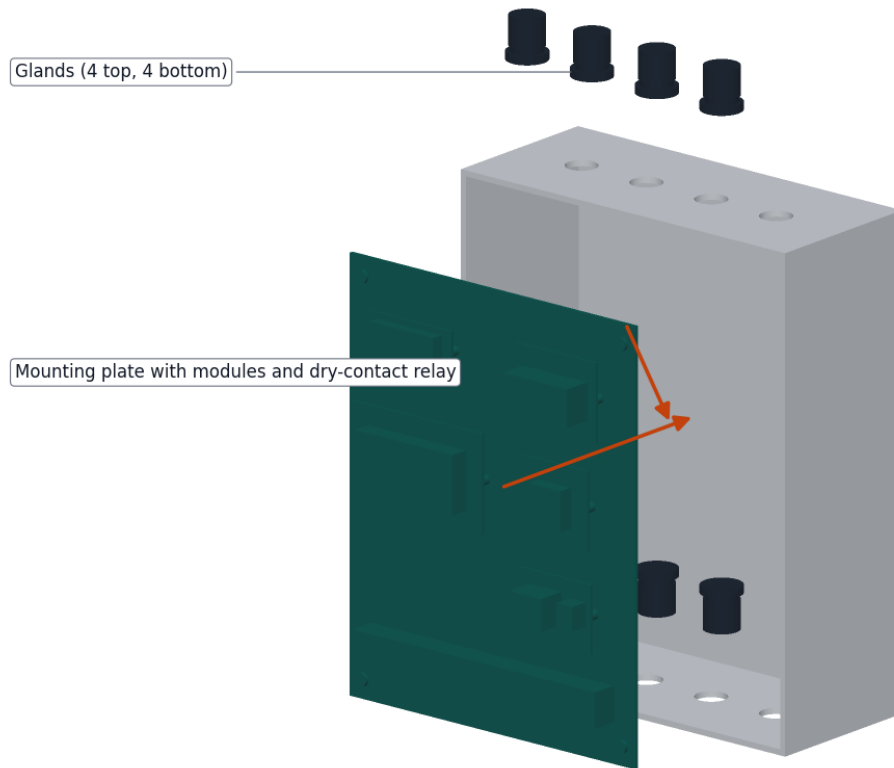
github.com/BoujeeEnjinia1701/h2guard

Find the apparatus centre on the ceiling with a plumb line and fix the flange there with the rod and its top locknut already on it (section 3.4), three M5 screws into ceiling plugs. Push the head up so the rod passes through the hole in its top and the top sits on the locknut, then fit the inner locknut through the open front and tighten it. The lid goes on after the cable is wired (step 17). **Hold point:** the sensor ports are 175 mm below the ceiling and the plumb line from the point midway between them lands on the apparatus centre.

Step 5: glands and mounting plate into the controller box

Step 5: glands and mounting plate into the controller box

Modules fitted to the plate on the bench first; plate on the four bosses with four M4 screws



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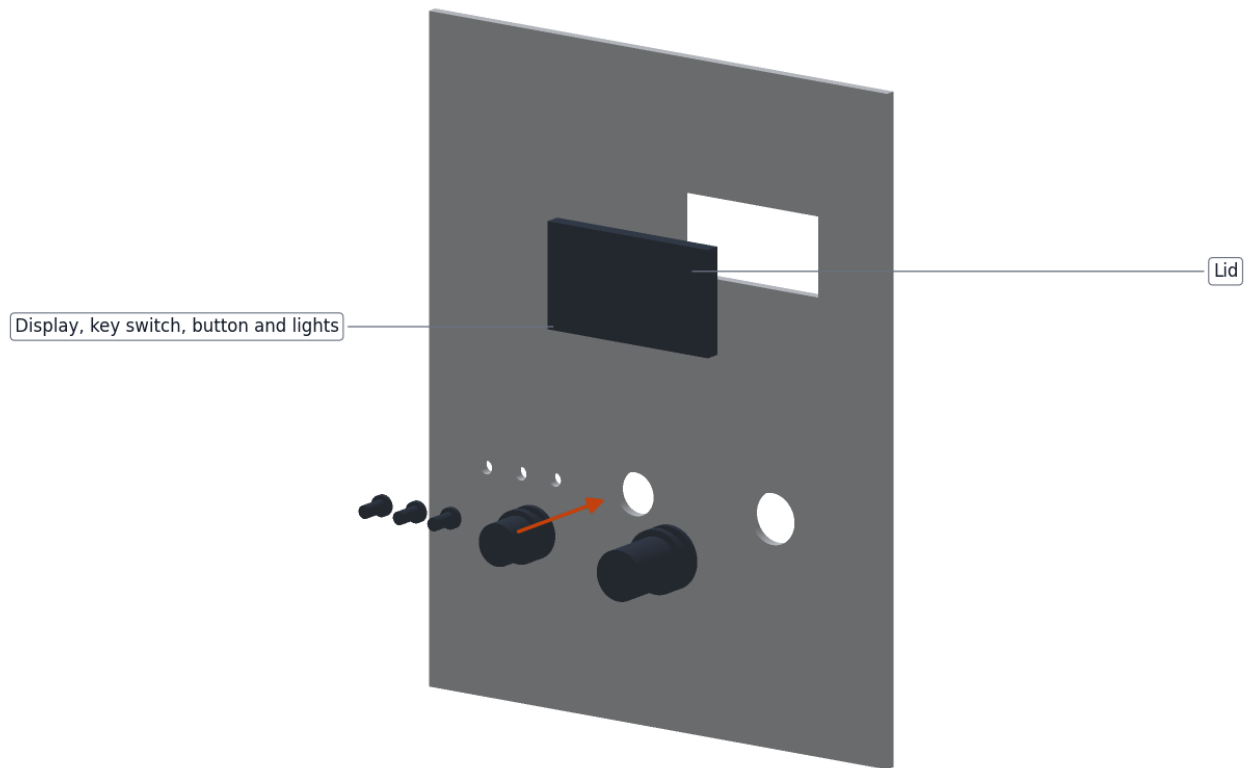
github.com/BoujeeEnjinia1701/h2guard

Fit the eight glands. With the modules and the dry-contact relay already on the plate and wired, set the plate on the bosses and fit four M4 self-tapping screws.

Step 6: front panel parts into the lid

Step 6: front panel parts into the lid

Seen from behind: display on four M2.5 screws; key switch, button and lights through their holes, nuts behind the lid



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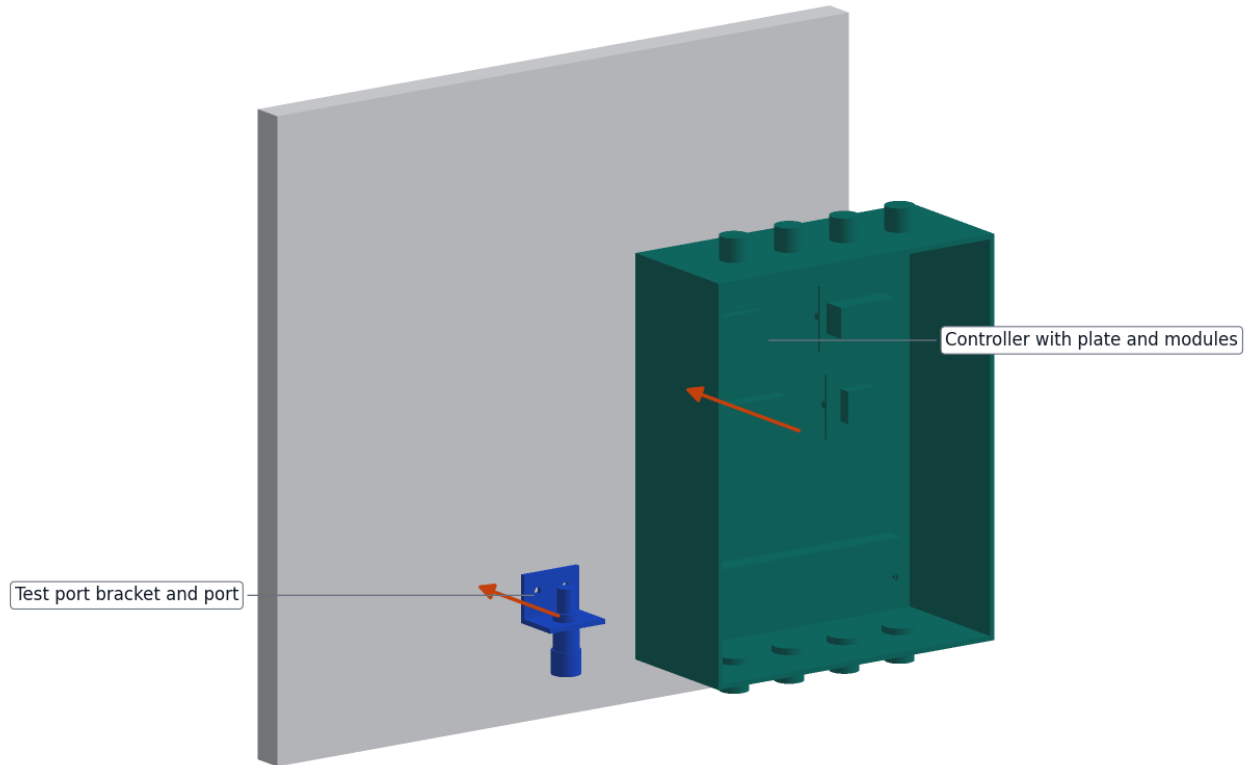
github.com/BoujeeEnjinia1701/h2guard

From behind the lid: the display on four M2.5 screws; the key switch, test button and lights through their holes, nuts behind the lid. Plug the ribbon cable into the display.

Step 7: controller and test port bracket onto the wall

Step 7: controller and test port bracket onto the wall

Seen from the front left. Controller top 1,475 mm above the floor, four screws; bracket 35 mm to its left, two M4 screws



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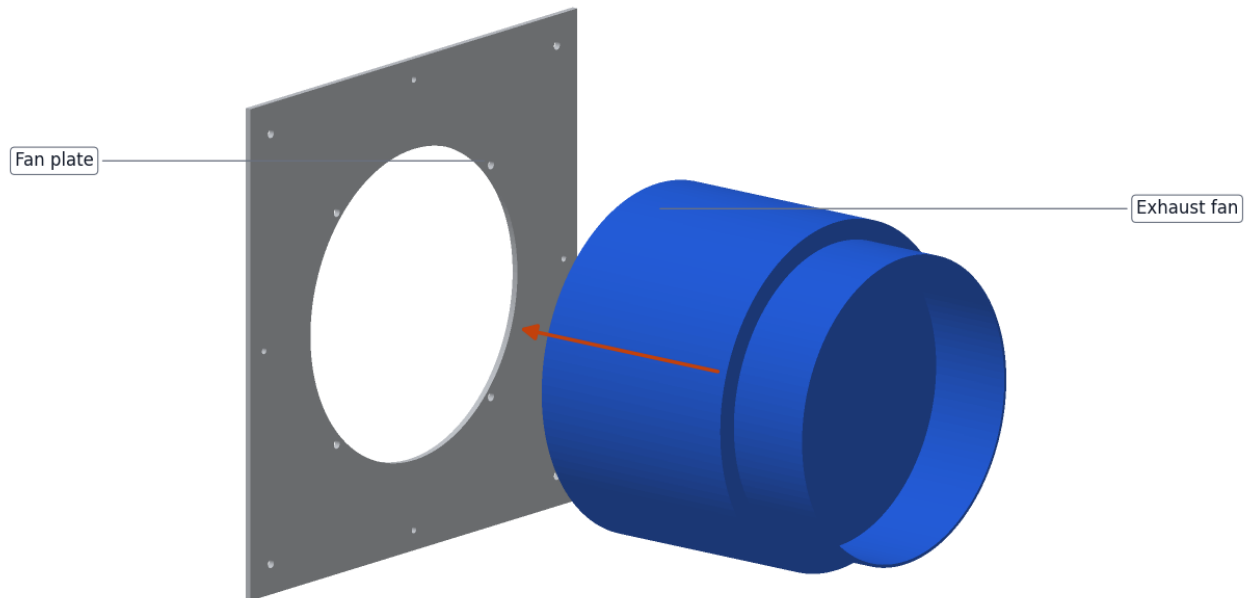
github.com/BoujeeEnjinia1701/h2guard

Controller box on the wall with its top 1,475 mm above the floor, four screws with plugs through the corner holes. Test port bracket 35 mm to its left, flat leg 1,250 mm up, two M4 screws with plugs. **Hold point:** the controller top is at least 1,000 mm below the ceiling.

Step 8: fan onto the fan plate

Step 8: fan onto the fan plate

On the bench. Fan inlet face flat on the back of the plate; four M4 countersunk screws from the front



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

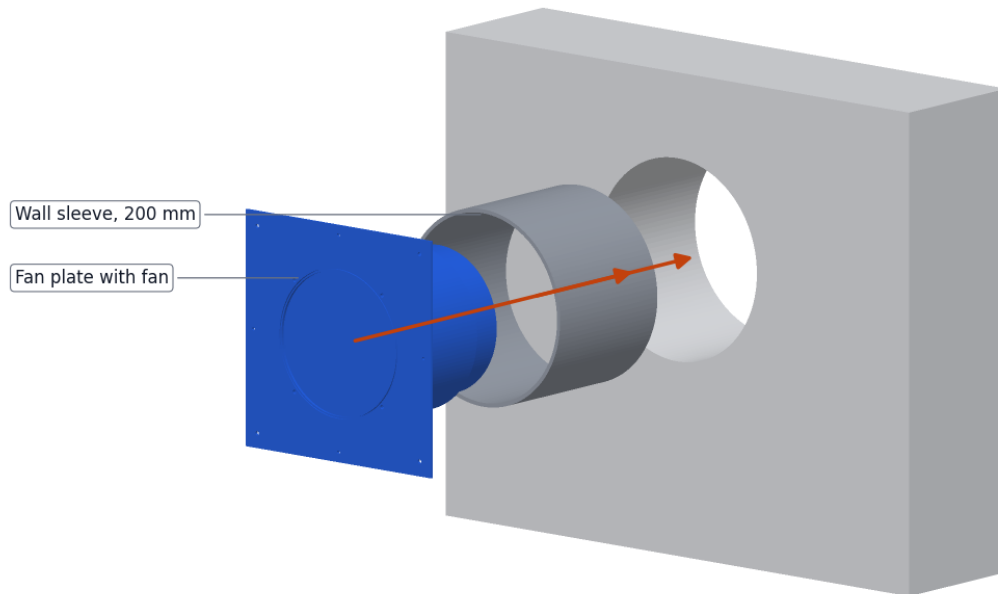
github.com/BoujeeEnjinia1701/h2guard

On the bench. The fan's inlet face flat on the back of the plate, four M4 countersunk screws from the front.

Step 9: sleeve and fan into the wall

Step 9: sleeve and fan into the wall

Sleeve flush with both wall faces; fan into the sleeve, plate flat on the wall, four screws with plugs



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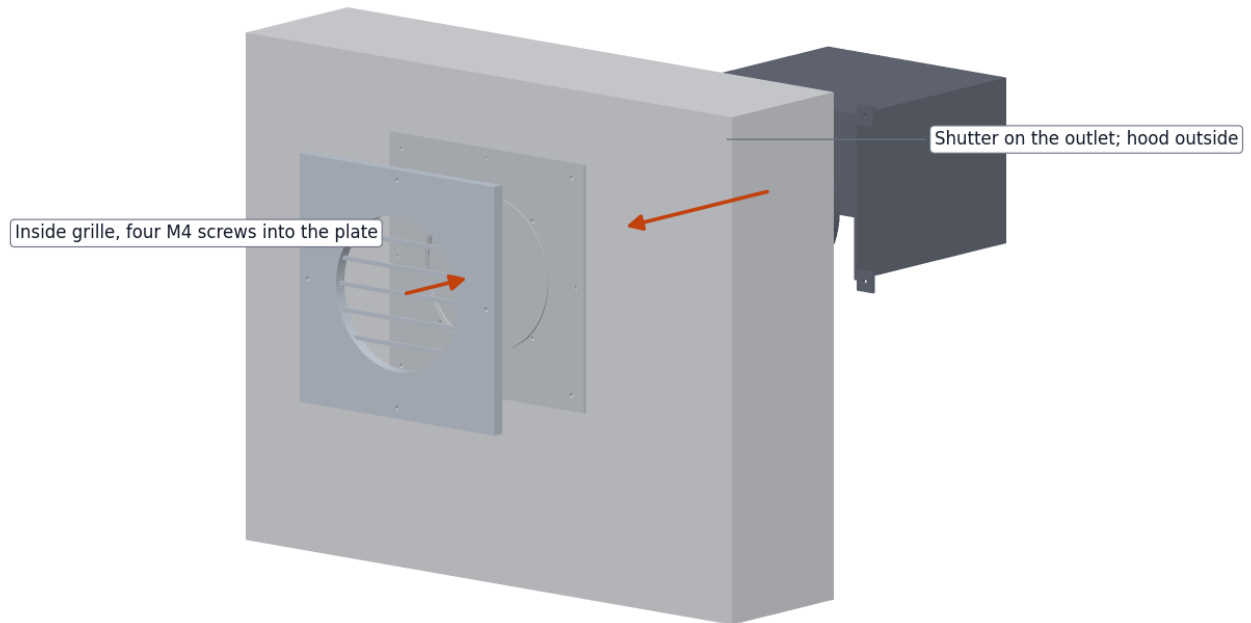
github.com/BoujeeEnjinia1701/h2guard

The builder's 206 mm opening is centred 2,300 mm above the floor. Push the 200 mm sleeve in flush with both wall faces. Slide the fan body into the sleeve and fix the plate flat on the wall with four countersunk screws and plugs.

Step 10: grille, shutter and weather hood

Step 10: grille, shutter and weather hood

Grille on the room side; shutter on the fan outlet and hood on four screws on the outside face



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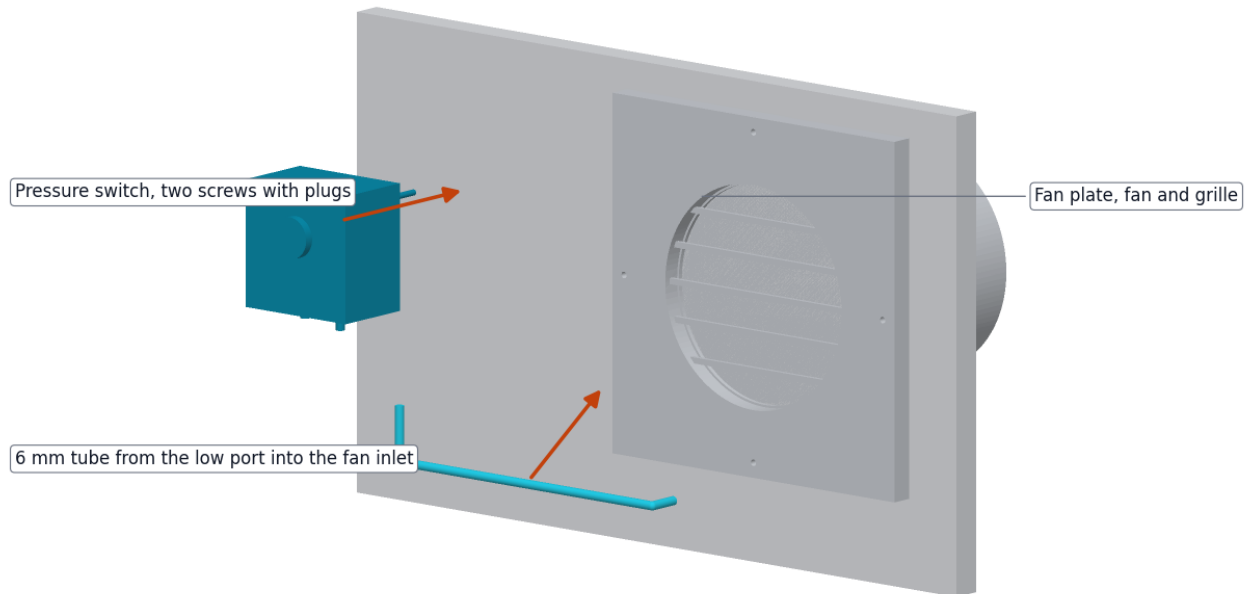
github.com/BoujeeEnjinia1701/h2guard

Inside grille onto the plate with four M4 screws. Outside, push the backdraft shutter onto the fan's outlet spigot, flap hanging closed, then fix the weather hood over it, open side down, with four screws and plugs.

Step 11: pressure switch and its tube

Step 11: pressure switch and its tube

Switch on the wall left of the fan, ports down; the tube's end sits just inside the grille, in the fan inlet



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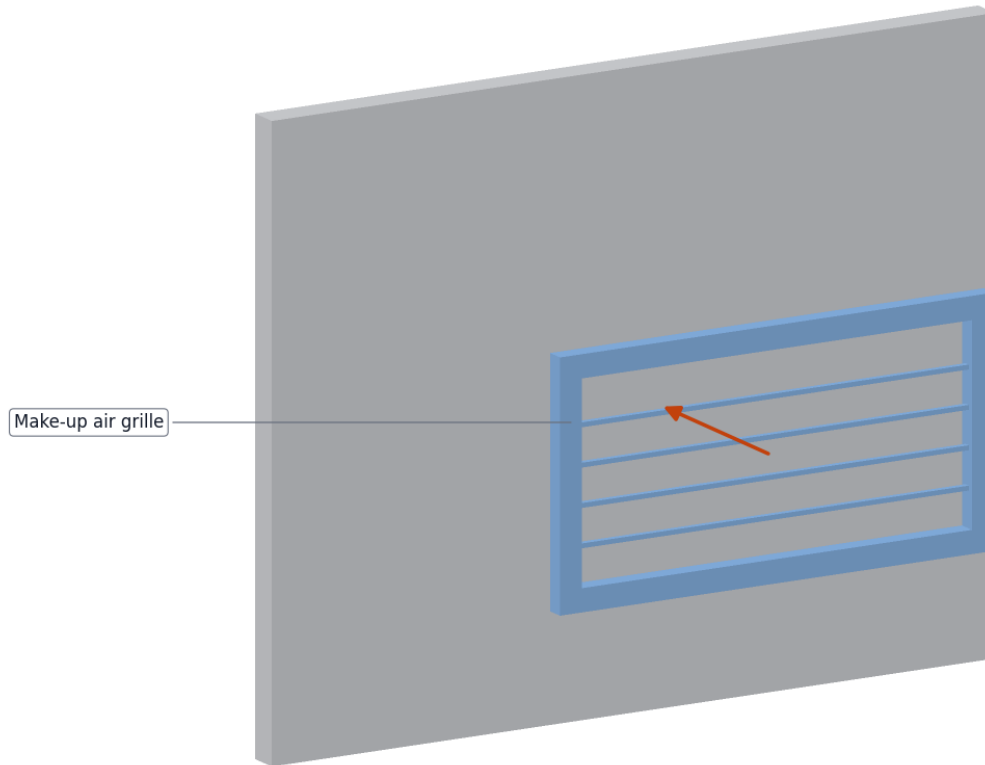
github.com/BoujeeEnjinia1701/h2guard

Switch on the wall left of the fan plate, ports down, two screws with plugs. Push the 6 mm tube onto the low port, run it down and across in front of the grille, and push its end through the grille opening, 62 mm below the fan's centre, until it sits just inside the grille. Leave the high port open. Set the switch to about 3 Pa.

Step 12: make-up air grille

Step 12: make-up air grille

Low on the far side wall over the builder's opening, mesh on; four screws with plugs



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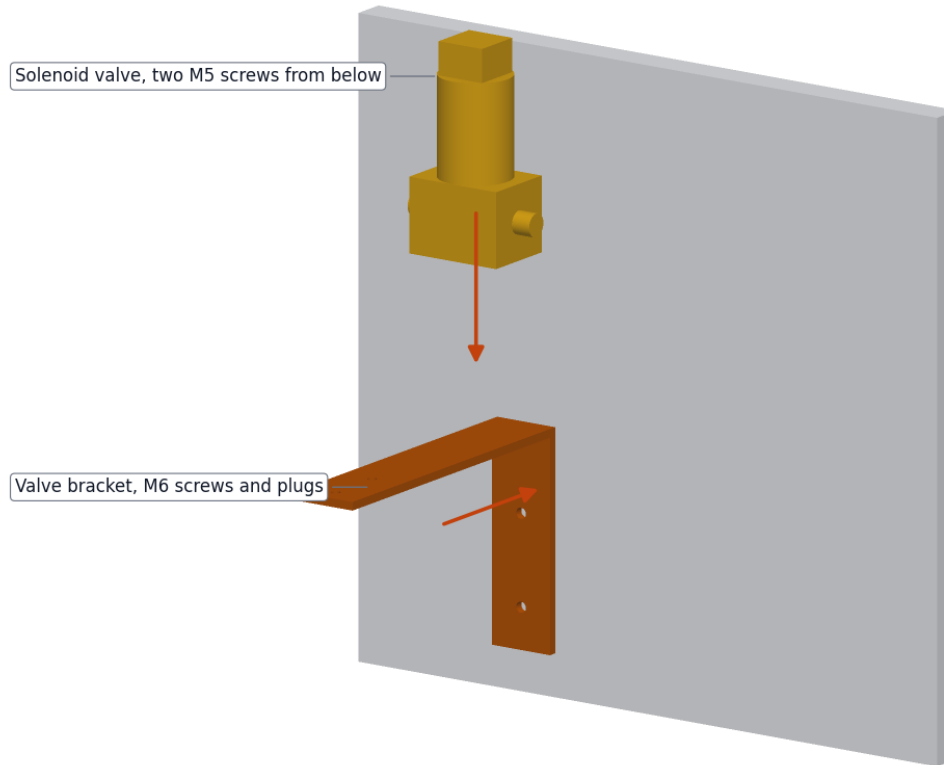
github.com/BoujeeEnjinia1701/h2guard

Low on the side wall opposite the fan, over the builder's opening, centred 200 mm above the floor, mesh fitted; four screws with plugs.

Step 13: valve bracket and valve

Step 13: valve bracket and valve

Arm level, 1,272 mm up; the gas fitter then connects the valve into the supply after the regulator



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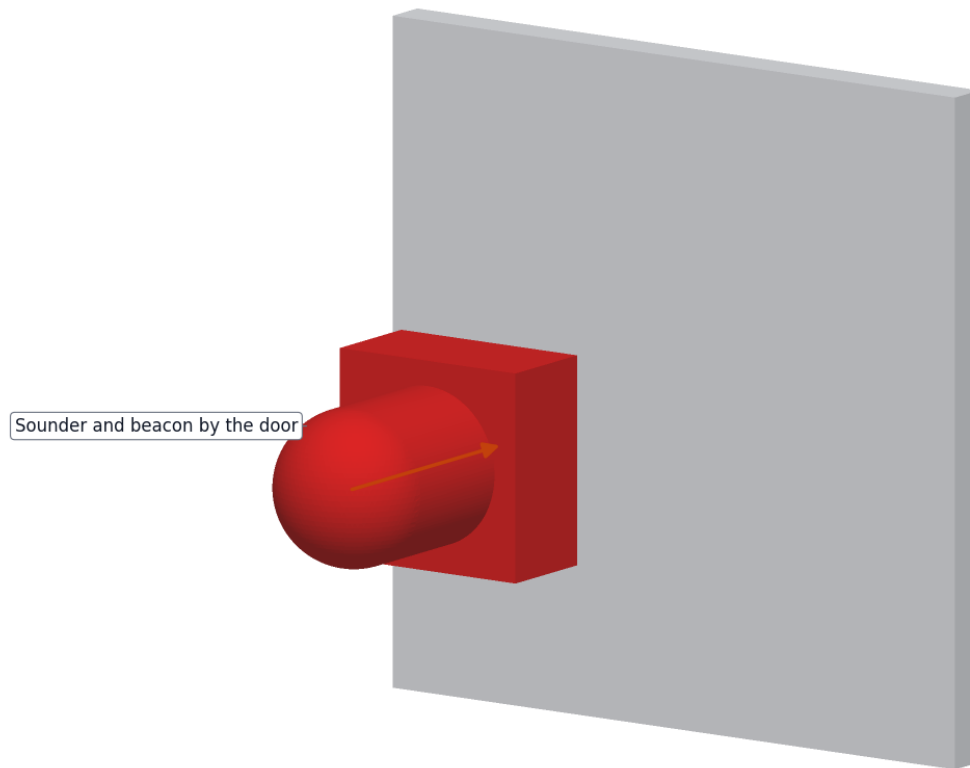
github.com/BoujeeEnjinia1701/h2guard

Bracket on the wall with the arm level and its top 1,277 mm above the floor, two M6 screws with plugs. Valve on the arm, two M5 screws from below. **Hold point:** the gas fitter connects the valve into the supply after the regulator and flow restrictor, and leak tests the joints with inert gas; no hydrogen is connected yet.

Step 14: sounder, beacon and power supply

Step 14: sounder, beacon and power supply

Beacon top 450 mm below the ceiling, two screws with plugs. The certified 24 V supply just stands on the floor by the outlet



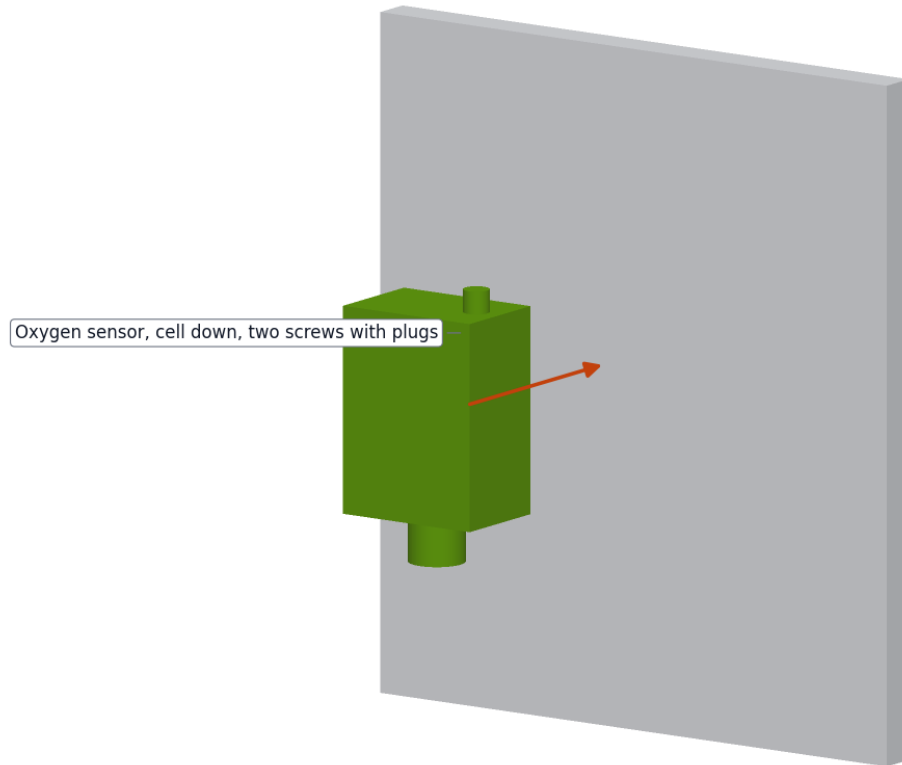
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Sounder and beacon by the door with its top 450 mm below the ceiling, two screws with plugs. Stand the 24 V supply on the floor by the outlet; do not plug it in yet.

Step 15: oxygen sensor (only where inert gas cylinders share the room)**Step 15: oxygen sensor (only where inert gas cylinders share the room)**

Cell at breathing height, about 1,480 mm above the floor; cable through the gland on top to the controller



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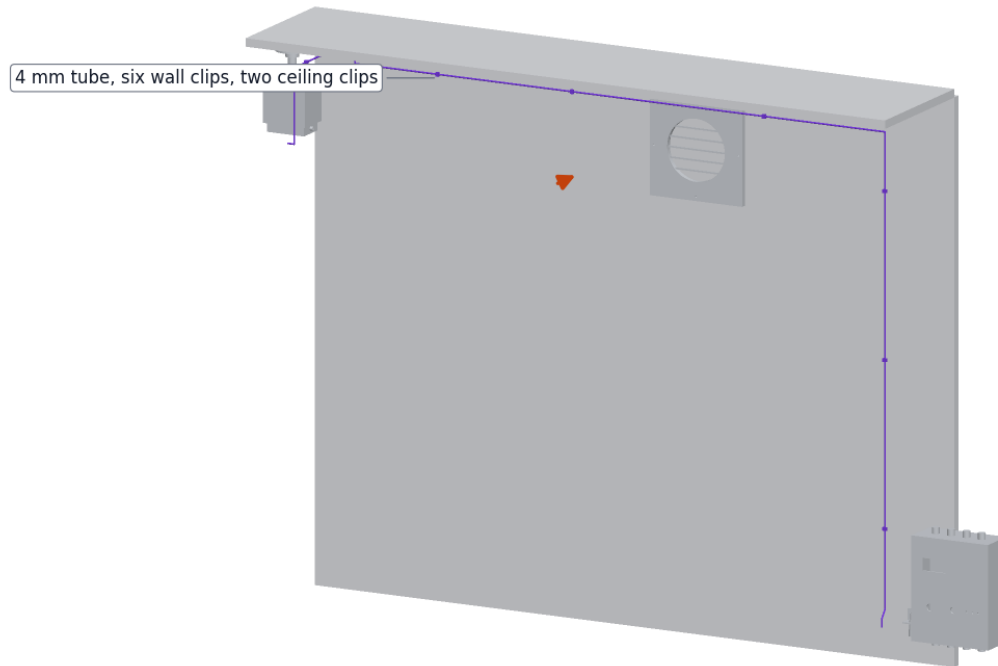
github.com/BoujeeEnjinia1701/h2guard

On the back wall, 450 mm left of the controller, with the cell facing down about 1,480 mm above the floor, at breathing height; two screws with plugs. Its cable runs from the gland on top down the wall to the left-hand bottom gland of the controller.

Step 16: bump test tube and clips

Step 16: bump test tube and clips

Tube from the top of the test port, up the wall, along under the ceiling, across the ceiling to the head and down into the cup fitting



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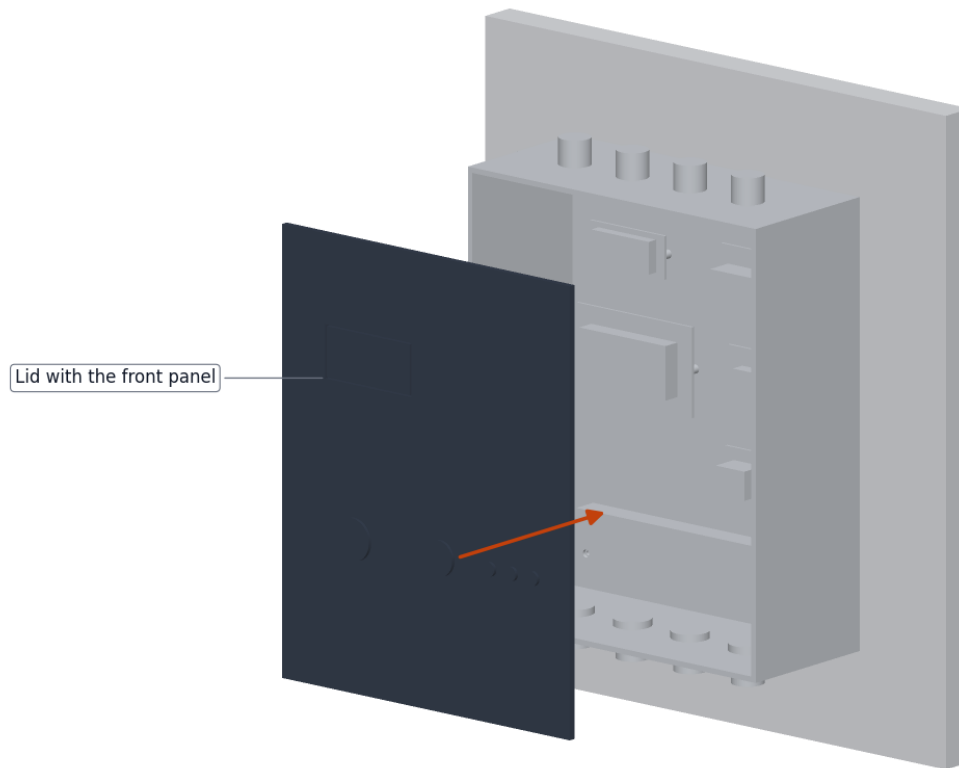
github.com/BoujeeEnjinia1701/h2guard

Push the 4 mm tube into the top of the test port, run it up the wall beside the controller and along the wall 60 mm under the ceiling, then up into the corner and out across the ceiling to the head, and down beside the head into the cup's fitting. Six saddle clips hold it on the wall and two on the ceiling; keep it clear of the fan grille, the drop rod and the cables.

Step 17: wire up and close the lids

Step 17: wire up and close the lids

Cables through the glands to the terminal strip, as the wiring diagram; lid on its gasket; same for the head lid



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github.com/BoujeeEnjinia1701/h2guard

Run the cables in surface conduit and through the glands to the terminal strip and the head's terminal, as Figure 13. Close the controller lid and the head lid on their gaskets. **Hold point:** safety stops S1 and S2 in section 6.

■ 5. First checks

These are the checks a TRL 4 test report would record; this plan only lists them. Requirement numbers are those of HGD-REQ-001.

Table 3. First checks.

Check	Requirement	How	Pass when
Mounting heights	R1, R15	Tape measure from the ceiling	Head ports 175 mm below the ceiling, give or take 10 mm; controller top at least 1,000 mm below
Head over the leak point	R1	Plumb line from the point midway between the head's ports	Lands on the apparatus centre, within 20 mm
Power draw	R11	Meter in the 24 V lead, normal state and with the test button held	Under 60 W at the peak (48.7 W estimated)
Fail-safe on power loss	R5	Unplug the 24 V lead with the valve open (inert gas in the line)	The valve closes; flow stops
Hardware trip	R6	With the microcontroller held in reset, feed the trip board's input a voltage equal to 25 % LFL	The series relay opens and the valve closes; only the key resets it
Sensor fault	R5	Open and then short the catalytic sensor input	Valve closed within 2 s; fault shown
Fan stop	R5, R7	Hold the fan still (supply off at the fan)	Valve closed within 2 s of the 10 s tachometer timeout
Airflow proving	R5, R7	With the fan running, cover the outside of the weather hood with a board	The pressure switch opens and the valve closes within 2 s of its 10 s timeout
Dry-contact output	R3, R5	Meter across the output terminal; press TEST, then unplug the 24 V lead	The contact opens each time
Oxygen sensor (if fitted)	Precis, decision 4	Read it in fresh air	About 20.9 % oxygen
Ventilation	R7	Vane anemometer across the inside grille at continuous and boost speed	At least 150 m ³ /h continuous and 300 m ³ /h on boost
Alarm	R10	Sound meter at 1 m from the sounder	85 dB(A) or more; beacon visible from the door
Bump test	R12	Certified 1 % hydrogen in air at 1 L/min into the test port (after stop S4)	The catalytic reading rises and the system trips within 120 s
Self-test	R12	Press TEST	Sounder, beacon, fan boost and valve close all run

■ 6. Safety stops

Stop at each point. Carry on only when everything listed is true.

- **S1. Before the 24 V supply is plugged in.** The supply is a certified unit with an undamaged lead. With it unplugged, no 24 V terminal reads short to 0 V. The valve coil wiring passes through the series relay and the driver switch. Every gland is tight and every lid closed.
- **S2. Before the sensors are powered.** The epoxy round the discs has cured for its full time with the head open; no silicone, flux cleaner or solvent has been used in or near the head.

- **S3. Before any gas reaches the valve.** A competent gas fitter has connected the valve after the regulator and flow restrictor and leak tested every joint with inert gas. The fault checks of section 5 (power loss, hardware trip, sensor fault, fan stop, airflow proving) all close the valve.
- **S4. Before the first bump test.** The test gas is a certified mixture of no more than 1 % hydrogen in air, in a disposable cylinder with its own regulator set to about 1 L/min. The fan is running. Someone stands by the controller with the key. Expect the system to trip; reset only when the reading is below 10 % LFL.
- **S5. Before hydrogen from the room's supply is turned on.** The hydrogen inventory in the room is within the 300 L limit (1 % of room volume at atmospheric pressure), or a flow restrictor limits a failure to 5 L/min. The bump test has passed that day. The room has a second, certified means of protection wherever codes or insurers require one. The head hangs on its drop rod with its ports over the apparatus (step 4). The pressure switch has proved that it closes the valve when the airflow stops. The valve is the hydrogen-rated valve, or a brass valve with its maker's hydrogen and seat leak ratings that has passed a seat leak proof test. If inert gas cylinders share the room, the oxygen sensor is fitted and reads about 20.9 % in fresh air. The set points are the stricter of the defaults, the national code, ISO 26142 and the host institution's safety office rules.
- **S6. Every teaching session.** Bump test before use and after any exposure to silicone or solvent vapour; check that the valve closes on TEST. A leaking valve seat is not detected by the system.

■ 7. Tools, skills and workspace

Tools. Bench drill or a drill in a stand; drills 2.5 to 12 mm; step drill to 20 mm; 152 mm hole saw (or a jigsaw with a metal blade); countersink; M4 and M5 taps and tap drills; hacksaw; bench vice with soft jaws and a radiused jaw for bending; flat and half-round files; deburring tool; scribe, square, steel rule and calipers; feeler gauges; 3D printer that prints PETG with a bed of at least 100 x 70 mm; soldering iron; ferrule crimper and wire strippers; multimeter; screwdrivers and nut drivers; masonry drill and wall plugs for the room; pipe wrench for the drop rod; spirit level; tape measure and plumb line. The fan and make-up air openings need a 206 mm core drill and a builder.

Skills. Basic metalwork (marking out, drilling, filing, tapping, bending flat bar), through-hole soldering, crimping, and reading a sensor datasheet and application note. All circuits are 24 V DC or lower; mains appears only inside the certified supply. The gas connection must be made and leak tested by a competent gas fitter.

Workspace. A bench about 1.2 x 0.6 m with the metalwork kept apart from the electronics; a ventilated place for the printer; a step ladder with a second person for work at the ceiling.

Personal protective equipment. Safety glasses for drilling, cutting and soldering; gloves for cut aluminium; dust mask and hearing protection when drilling masonry; no gloves near a turning drill.

■ 8. Where the numbers come from

- Model and constructability checks: `cad/src/model.py` (`python cad/src/model.py --check` , 182 checks); STEP and STL exports in `cad/step/` and `cad/stl/` .
- Pictures: `cad/src/build_plan_media.py` , using `.kit/build_views.py` ; written to `docs/05-build-plan/` and `cad/drawings/HGD-DWG-101` to `HGD-DWG-110` .
- General arrangement: `cad/drawings/HGD-DWG-001.pdf` , Rev P5.
- Calculations: `docs/04-calcs/01-sizing.md` (HGD-CAL-001 v0.4) and `docs/04-calcs/sizing.py` ; head over the plume [E7], arrestor lag [F1], fan delivery [G3], pressure switch [G6], power [H2], bump test [J1], installation [M1], cost [N1].
- Bill of materials: `bom/bom.csv` .

- Decisions: docs/decisions/0003-design-for-construction.md (HGD-DDR-003), with HGD-DDR-001 and HGD-DDR-002.
- Requirements: docs/03-requirements.md (HGD-REQ-001 v0.7).